

EXB-01 · EXAM BLUEPRINT

Examination blueprint — PCL-AI

What the examination assesses, in what proportion, in what format — and what is not yet fixed

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

This is the published specification for the PCI AI Project Controls Leader (PCL-AI) examination: the three assessment groups and their 40/40/20 weighting, all thirteen domains and sixty-one knowledge areas that may be sampled, the item format, the cognitive levels, the scoring basis and the duration. It also states plainly what has not yet been fixed — the number of items and the definitive passing standard — and how each will be settled. Read it as the contract between the candidate and the examination: study against it, audit yourself against it, and hold the Institute to it. It covers the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

— About this document

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Examination blueprint — PCL-AI

In one paragraph. This is the published specification for the PCI AI Project Controls Leader (PCL-AI) examination: the three assessment groups and their 40/40/20 weighting, all thirteen domains and sixty-one knowledge areas that may be sampled, the item format, the cognitive levels, the scoring basis and the duration. It also states plainly what has not yet been fixed — the number of items and the definitive passing standard — and how each will be settled. Read it as the contract between the candidate and the examination: study against it, audit yourself against it, and hold the Institute to it. It covers the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

Who this is for. Candidates preparing for the PCL-AI; training providers building courses against it; employers deciding what a PCL-AI holder was actually assessed on; and assessment professionals auditing whether the Institute's claims about its examination are supported by a published specification.

— 1. What a blueprint commits us to

A blueprint is not a syllabus with a marketing wrapper. It is the document that makes an examination auditable, and it does three specific things.

It fixes the sampling frame. Every item on every form is drawn from the domains and knowledge areas set out in §5 of this document, at the group weights set out in §4. Content that is not in the frame is not examinable. A candidate who studies the frame has studied the examination; there are no surprise topics and no unpublished content.

It makes forms comparable. Because every form is assembled to the same specification, a pass earned on one form means what a pass earned on another form means. Without a blueprint, "the examination" is whatever questions happened to be selected that month, and comparability across sittings is an assertion rather than a property of the design.

It transfers the burden of proof onto the Institute. Publishing the specification means anyone can check whether our examination does what we say it does. That is uncomfortable by design. An assessment that cannot be inspected cannot be trusted, and the Institute's position is that a certification body which will not publish its blueprint is asking to be taken on faith.

1.1 What this blueprint governs, and what it does not

This blueprint governs	The published certification policies govern
Which domains and knowledge areas may be sampled	Eligibility, application and verification of experience
The group weightings applied to every form	Fees, booking, rescheduling and cancellation
Item format and the cognitive levels items are written to	Identification, proctoring arrangements and accommodations
The scoring basis and how the passing standard is set	Misconduct, appeals, complaints and retakes
The item development, review and retirement process	Recertification, continuing professional development and use of the credential

Where this blueprint and a published policy appear to conflict, the published policy prevails and the conflict is a defect in this document that the Institute wants reported. For the policy side of the table, see [CER-01 – Certification handbook – master](#).

— 2. The credential this covers — and the two it does not

This blueprint applies to the **PCL-AI (PCI AI Project Controls Leader)** only.

The Institute certifies three credentials at Leader level: PCL-AI, PFL-AI (PCI AI Project Finance Leader) and PML-AI (PCI Project Management Leader – AI). **No examination blueprint exists yet for PFL-AI or PML-AI.** There are no published weightings, no published domain structure and no item counts for those two examinations, and nothing in this document should be read as covering them. When their blueprints are developed they will be published in this series under their own identifiers, built by the same process, and subject to their own job-task analysis.

A fourth offering, the **AI in Project Controls — Specialist Certificate (AIPC)**, is a certificate rather than one of the three credentials. It is open-entry, drawn from Domain 13 of the body of knowledge below, and assessed by scenario items plus an applied AI task. It is not governed by this blueprint.

An earlier edition of this blueprint was published under a credential code that has since been retired. Where legacy study material carries that older code, the content it describes is now the PCL-AI; the framework did not change, the name did.

— 3. The examination at a glance

Element	Specification
Duration	90 minutes
Item format	Four-option multiple choice, single best answer
Number of scored items	[CONFIRM: number of scored items — to be set by the job-task analysis] — see §9.1
Cognitive levels	Recall, Application, Analysis — application-weighted (see EXB-03)
Content sampling	Thirteen domains, sixty-one knowledge areas, in three groups weighted 40/40/20
Scoring	Criterion-referenced, not graded on a curve
Configured pass mark	65 per cent — the definitive standard will be set by a modified-Angoff standard-setting study (see §9.2 and EXB-05)
Formula sheet	None provided in the examination (see §7.2)
Result	Issued immediately on submission, with a domain-level breakdown
Interface languages	English, Arabic, Spanish, French, Korean, Chinese, Russian
Proctoring	Live proctoring with identity verification, room scan and second-monitor blocking (see EXB-07)
Launch window	Opens 15 minutes before the booked time, with a 30-minute grace period
Certificate validity	3 years, maintained through continuing professional development with a mandatory AI-currency component

Booking, rescheduling, attempt allowances and fees are governed by the published policies, not by this blueprint; see [CER-01](#).

— 4. The three groups and the 40/40/20 weighting

Every form is assembled so that the scored content divides across three groups in fixed proportions.

Group	Domains	Knowledge areas	Weight
Project accounting & finance	D1–D4	19	40 %
Project management principles	D5–D12	35	40 %
AI knowledge & practical approach	D13	7	20 %
Total	13 domains	61	100 %

Two things follow from that table, and both matter more than they look.

Weights attach to groups, not to individual domains. The Institute publishes no per-domain percentage and no per-domain passing requirement. Within a group, coverage is spread across its member domains and their knowledge areas; the blueprint constrains the group totals, and the item selection rules constrain the spread within a group so that no domain can be omitted from a form or allowed to dominate it.

The three groups are not equal in size, and the weighting is not proportional to size. Nineteen knowledge areas carry 40 per cent; thirty-five knowledge areas carry the same 40 per cent; seven knowledge areas carry 20 per cent. That is a deliberate measurement decision about what the credential attests, and it has consequences a candidate can compute for their own revision plan. The full defence of the weighting, and the arithmetic of what it does to a score, is in [EXB-02 – Domain weightings and the 40/40/20 rule](#).

— 5. The syllabus: thirteen domains, sixty-one knowledge areas

This is the sampling frame in full. Every knowledge area listed is examinable. The numbering convention is **Domain.KnowledgeArea.Topic** — so 6.2 is the second knowledge area of Domain 6, and 6.2.1 a topic within it. Items are tagged to that level so that domain-level reporting on a result is a fact about the form, not an estimate.

5.1 Group 1 — Project accounting & finance (40 %)

Nineteen knowledge areas across four domains. This group exists because a controls professional who cannot read the accounts cannot reconcile the cost report to them, and a cost report that does not reconcile is an opinion.

Domain	Knowledge areas
D1 — Foundations of Accounting for Project Controls (5)	1.1 The accounting model · 1.2 Components of the financial statements · 1.3 Accrual accounting and the matching concept · 1.4 Cost provisions and cost accruals · 1.5 Chart of accounts and cost coding for projects
D2 — Financial Reporting & the Standards (5)	2.1 The reporting framework · 2.2 IFRS 15 Revenue from Contracts with Customers · 2.3 Revenue recognition beyond the basics · 2.4 Other relevant standards for project controls · 2.5 Management reporting versus statutory reporting
D3 — Budgeting & Forecasting (5)	3.1 Budgeting fundamentals · 3.2 Cost estimation · 3.3 The time-phased budget and cost baseline (planned value) · 3.4 Forecasting · 3.5 Cash-flow forecasting
D4 — Performance Management, Variance Analysis & Management Reporting (4)	4.1 Performance management principles · 4.2 Variance analysis · 4.3 Management reporting · 4.4 Data visualisation and storytelling for controls

Standards are examined at the level a controls professional must actually work with them: the principle, the recognition test, the consequence for the project position and the presentation. Items describe requirements in the Institute's own words. Protected standard text, tables and diagrams are never reproduced in study material or in an item.

5.2 Group 2 — Project management principles (40 %)

Thirty-five knowledge areas across eight domains. This is the discipline itself — cost, earned value, commercial, lifecycle, adaptive delivery, schedule, process cycles and risk.

Domain	Knowledge areas
D5 — Cost Management & Cost Control (4)	5.1 The cost management framework · 5.2 The cost control cycle · 5.3 Cost breakdown and control accounts · 5.4 Change control and cost impact
D6 — Earned Value Management & Forecasting (EVM/EAC) (4)	6.1 EVM fundamentals · 6.2 Variances and performance indices · 6.3 Forecasting with EVM: the estimate-at-completion family · 6.4 Integrating cost and schedule; limitations; earned schedule
D7 — Contracts, Commercial Management, BoQ, Invoicing & Revenue (5)	7.1 Types of contract · 7.2 Contract management · 7.3 Bills of quantities · 7.4 Invoicing and applications for payment · 7.5 Revenue recognition in the commercial cycle
D8 — Project Management Lifecycle (6)	8.1 Initiating · 8.2 Planning · 8.3 Executing · 8.4 Monitoring and controlling · 8.5 Closing · 8.6 Development approaches: predictive, iterative, incremental and adaptive
D9 — Agile, Scrum & Adaptive Delivery for Project Controls (6)	9.1 Agile foundations · 9.2 The Scrum framework in depth · 9.3 Backlogs, estimation and agile metrics · 9.4 Kanban, Lean and scaling · 9.5 Agile cost control, forecasting and earned value · 9.6 Hybrid delivery and agile governance
D10 — Project Scheduling (4)	10.1 Schedule development · 10.2 Network analysis and the critical path method · 10.3 Schedule compression and resourcing · 10.4 Progress measurement and schedule control
D11 — Business Process Cycles (3)	11.1 Order-to-cash · 11.2 Procure-to-pay · 11.3 Internal control and segregation of duties
D12 — Risk Management for Project Controls (3)	12.1 The risk framework · 12.2 The risk process · 12.3 Contingency and management reserve

Abbreviations used above and throughout: **BoQ** — bills of quantities; **EVM** — earned value management; **EAC** — estimate at completion.

5.3 Group 3 — AI knowledge & practical approach (20 %)

Seven knowledge areas in one domain, at a fifth of the examination.

Domain	Knowledge areas
D13 — AI for Project Controls & PM: Concepts, Tools & Practice (7)	13.1 AI foundations for professionals · 13.2 Data: the fuel · 13.3 Prompting and working with generative AI · 13.4 AI tool categories for project controls and project management · 13.5 AI applied across the project-controls lifecycle · 13.6 Governance, ethics, risk and assurance of AI · 13.7 Building an AI-augmented project-controls capability

The domain is vendor-neutral. Items test what a technique does, what it needs, where it fails and who is accountable for its output — never the interface of a particular product. A candidate who has learned one tool's menus has not prepared for this group.

— 6. How AI is examined

AI is not confined to Group 3. Two things are true at once, and candidates routinely prepare for only one of them.

Domain 13 carries the systematic treatment, at the published 20 per cent group weight: what the techniques are, what data they need, how they are governed, how they are assured, and how a capability is built. That is examined on its own terms.

Governed-AI judgement is also examinable inside the other two groups. Every domain in the body of knowledge carries embedded coverage of AI as it applies to that domain — automated cost coding in D1, predictive estimate-at-completion in D6, exception detection in D11, risk scoring in D12. An item in Group 1 or Group 2 may therefore put an AI-generated output in front of the candidate and ask what a competent professional does with it. Such an item is scored against the domain of its subject matter, not against Domain 13; a question about an AI-generated cost forecast is a cost-forecasting question.

The Institute's governing position runs through all of it: **AI proposes; the professional disposes.** No item will ever reward a candidate for accepting a machine output that is not explainable, validated and owned by a competent human. Several will penalise it.

— 7. Item format

7.1 Four options, one best answer

Every item is a four-option multiple-choice question with a single best answer. Most are scenario-based: a situation from controls work, the data needed to judge it, and four options that a competent practitioner would recognise as arguable. The task is usually not to spot a wrong number but to decide what to do next.

The format's strengths and its real limitations — what a single-best-answer item can and cannot measure, and why the Institute has chosen it anyway — are set out in [EXB-03 – Cognitive levels and item types](#). Worked examples with a rationale for every option are in [EXB-08 – Sample items with rationale](#).

7.2 Calculation items and the formula policy

Calculation items supply the data. They do not supply the method.

No formula sheet is provided in the examination. Candidates are expected to know the core relationships of the body of knowledge — the accounting equation, the earned value family and its variances and indices, the estimate-at-completion forms, variance decomposition, expected value — and, more importantly, to select the right one for the situation. Choosing between two defensible estimate-at-completion methods on the evidence in front of you is a substantial part of what the credential attests; handing the candidate a menu of formulae removes the assessment along with the memory load.

The Institute publishes a master formula sheet as **study material**. It is a preparation aid, not an examination aid.

— 8. Scoring

Scoring is **criterion-referenced, not graded on a curve**. Everyone who demonstrates the required standard passes, however many others do. There is no quota, no fixed proportion who succeed and no ranking against other candidates. A candidate's result reflects their performance against a published standard and nothing else — which is why the Institute publishes no pass rate as a target and would regard a pass rate managed towards a number as a governance failure.

Results are issued immediately on submission with a domain-level breakdown, so an unsuccessful candidate receives a map for targeted preparation rather than a bare verdict. A result placed under integrity review is a deliberate exception: it shows no score and no pass or fail outcome until the review concludes. That rule, and why it exists, is set out in [EXB-07 – Examination security and integrity](#) §5.

— 9. What is deliberately not yet fixed

The Institute is founding-stage. Two elements of this specification are genuinely open, and this document states them as open rather than inventing a figure that would later have to be quietly changed.

9.1 The number of items

No item count exists. There is no published number, no configured number and no working assumption being held back. **Item counts will be confirmed by a formal job-task analysis** — a structured study, with practitioners, of the tasks the credential attests and their criticality and frequency. That study determines how many items each group needs to measure its content reliably, and therefore what the total is. Setting the count first and finding a justification afterwards is the failure mode; see [EXB-06 – Job-task analysis: where the blueprint gets its authority](#).

Anyone quoting an item count for this examination is quoting something the Institute has not published.

9.2 The definitive passing standard

65 per cent is the platform's configured pass mark, and it is what the examination applies today. **The definitive standard will be set by a modified-Angoff standard-setting study** — a documented, panel-based judgement of what a minimally competent PCL-AI holder must be able to do, expressed as a cut score on the assembled form. Both statements are true and both are published, because a configured value and a validated standard are different things, and a candidate is entitled to know which one they are being measured against.

If the study sets the standard somewhere other than 65 per cent, the Institute will say so, publish the study's method and panel composition, and apply the new standard from a stated date — not retrospectively. The method, the panel and the arithmetic are set out in [EXB-05 – Standard setting: modified Angoff and the pass mark](#).

9.3 What is not open

To be equally clear about the other direction: the duration, the item format, the cognitive levels, the group weighting, the domain and knowledge-area structure, the criterion-referenced scoring basis, the absence of a formula sheet and the three-year validity are all settled and published above. They are not provisional, and a change to any of them would be a new version of this blueprint under §12.

— 10. What this blueprint does not do

Honesty about a specification's limits is part of its value.

It sets no per-domain minimum. Weights attach to the three groups. There is no published per-domain percentage and no requirement to reach a threshold in each domain separately. The domain breakdown on a result is diagnostic information, not a set of sub-hurdles.

It is not an item list. The blueprint governs sampling. The items on any live form are secured content and are never published. The sample items in [EXB-08](#) are written for publication and are held apart from the live bank; see [EXB-07](#) §2.

It guarantees nothing about outcomes. Studying the blueprint is necessary and not sufficient. No preparation, from any provider or none, guarantees a pass. The certification decision is independent of whatever training a candidate did or did not take, and the Institute makes no representation about pass rates, salaries or employment outcomes.

It claims no recognition the Institute does not hold. The certification framework is developed with reference to ISO/IEC 17024 personnel-certification principles. PCI is **not accredited** by ANAB, IAS or any other ISO/IEC 17024 accreditation body, and this blueprint carries no governmental approval or university equivalence. Where our status is relevant, we state it plainly.

It is not the policy of record. Fees, booking windows, retakes, accommodations, conduct and appeals are governed by the published policies summarised in [CER-01](#).

— 11. Using the blueprint as a revision plan

The weighting is a study budget, and most candidates spend it badly by revising what they enjoy.

Rate your own confidence, honestly, against each of the thirteen domains in §5 — not against the three group headings, which are too coarse to act on. Then allocate preparation time against the published group weights and your gaps, in that order. Three patterns recur.

Candidates from a finance or accounting background typically under-prepare Domains 8 to 10 — lifecycle, adaptive delivery and schedule logic. They can read a set of accounts and cannot read a network.

Candidates from a planning background typically under-prepare Domains 1 to 4. The 40 per cent accounting and finance group is the largest single gap among practitioners who came up through scheduling, and it is the one most often discovered too late.

Almost every candidate under-prepares Domain 13 relative to its weight, because it feels peripheral to the day job. At the published weighting, one scored item in five sits in the AI group before any embedded coverage elsewhere is counted. Treating it as a topic to skim is the most expensive study decision available.

Practise across domains rather than within them. Examination scenarios cross domain boundaries deliberately — a schedule slip has cost, commercial, risk and reporting consequences, and an item may ask about any of them — so revising each domain in isolation prepares you for a form that does not exist.

— 12. Versioning and change

This blueprint is maintained under PCI governance alongside the body of knowledge.

A **minor version** corrects, clarifies or adds explanation without changing what is examined or how it is weighted. A **major version** changes the sampling frame, the weighting, the format, the cognitive levels or the scoring basis, and is published with a stated effective date and a notice period.

Candidates are assessed only against the edition of the blueprint in force for their sitting.

A change published after a candidate books does not apply to that sitting unless it is applied in the candidate's favour. The effective date of this edition and the date of the first live sitting are [\[CONFIRM: effective date of this blueprint edition and the date of the first live sitting\]](#).

— 13. Candidate self-audit checklist

Take this into your revision planning, not into the examination.

- [] I can name all thirteen domains and say which of the three groups each belongs to.
- [] I have rated my confidence against each of the sixty-one knowledge areas in §5, not just the domains.
- [] My revision time is allocated 40/40/20 against the groups, adjusted for my weakest areas — not for my favourites.
- [] I can compute the earned value family, its variances and indices, and at least three estimate-at-completion forms without a reference sheet, and say when each is appropriate.
- [] I can read a contract type and state who carries cost-overflow risk on defined scope.
- [] I can build and read a critical path, compute total float, and say what happens when float is exceeded.
- [] I can explain how revenue recognition, billing and the cost report relate to one another on a single contract.
- [] I can describe what a machine-learning model needs from project data before it is any use, and what happens when it does not get it.
- [] For any AI-generated output, I can state how I would establish that it is explainable, validated and owned.
- [] I have practised at least one scenario that crosses three domains and answered it under time pressure.
- [] I have read [EXB-07](#) and know what conduct will put a result under integrity review.

— Related

- [EXB-02 – Domain weightings and the 40/40/20 rule](#) — why the weighting is what it is, and exactly what a group weight does to a score
- [EXB-03 – Cognitive levels and item types](#) — what Recall, Application and Analysis mean here, and the limits of the four-option format
- [EXB-06 – Job-task analysis: where the blueprint gets its authority](#) — the study that will fix the item counts this document leaves open
- [EXB-08 – Sample items with rationale](#) — the format worked in full, with every option explained
- [CER-01 – Certification handbook – master](#) — eligibility, booking, conduct, results and appeals: everything §1.1 puts outside this blueprint

- [PCB-02 – The thirteen domains at a glance](#) — the same structure read as a body of knowledge rather than as a sampling frame

— Sources and standards

The domain and knowledge-area structure in §5 is taken from the PCI Body of Knowledge ([docs/bok/](#)), which is the source of truth for content; this document reproduces its structure, not its text. The examination parameters in §3 are the Institute's configured platform settings as verified in August 2026.

Standards named in the syllabus — IFRS and IAS reporting standards, the AACE International total cost management framework, PMBOK, ISO 31000, the Agile Manifesto and the Scrum Guide, and ISO/IEC 17024 — are referenced by name and explained in the Institute's own words throughout the body of knowledge. No protected text, table, diagram or item bank from any of them is reproduced.

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Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

EXB-02 · EXAM BLUEPRINT

Domain weightings and the 40/40/20 rule

Why the PCL-AI examination is weighted the way it is, and exactly what a group weight does to a score

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

The PCL-AI examination weights three groups at 40 per cent for project accounting and finance, 40 per cent for project management principles and 20 per cent for AI knowledge and practical approach. This document defends that weighting as a measurement decision rather than a preference, shows the arithmetic of how a group weight converts into a score contribution, works out what a weak group actually costs a candidate, and states plainly what the weighting does not do — including the fact that it sets no per-domain and no per-group hurdle. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

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Domain weightings and the 40/40/20 rule

In one paragraph. The PCL-AI examination weights three groups at 40 per cent for project accounting and finance, 40 per cent for project management principles and 20 per cent for AI knowledge and practical approach. This document defends that weighting as a measurement decision rather than a preference, shows the arithmetic of how a group weight converts into a score contribution, works out what a weak group actually costs a candidate, and states plainly what the weighting does not do — including the fact that it sets no per-domain and no per-group hurdle. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

Who this is for. Candidates planning revision time; training providers allocating course hours; assessment professionals auditing whether the weighting is defensible; and employers asking what proportion of a PCL-AI holder's assessment was actually about their own discipline.

— 1. A weighting is a claim, not a preference

Every weighted examination makes a public assertion: this is how much of the competence we are certifying sits in that part of the discipline. It is the most consequential single number in an examination specification, because it determines what a candidate studies, what a training provider teaches, and what a pass actually attests.

Most certification bodies publish a weighting and stop. The number arrives without a defence, and a reader cannot tell whether it came from a job-task analysis, a committee vote, or the relative enthusiasm of whoever wrote the syllabus. The Institute's position is that an undefended weighting is an unaudited claim, so this document does three things: it states the arithmetic consequences precisely, it gives the grounds for the decision, and it names the point at which the decision will be tested against evidence rather than argument.

The weighting itself is set out in [EXB-01 – Examination blueprint – PCL-AI §4](#), which is the authoritative statement. This document explains and defends it.

— 2. The weighting, and what is unusual about it

Group	Domains	Knowledge areas	Share of knowledge areas	Weight
Project accounting & finance	D1–D4	19	31.1 %	40 %
Project management principles	D5–D12	35	57.4 %	40 %
AI knowledge & practical approach	D13	7	11.5 %	20 %
Total	13	61	100 %	100 %

Shares of knowledge areas are $19 \div 61$, $35 \div 61$ and $7 \div 61$, rounded to one decimal place.

The weighting is deliberately **not** proportional to the size of the content. Thirty-five knowledge areas of project management principles carry the same weight as nineteen knowledge areas of accounting and finance, and seven knowledge areas of AI carry half as much as either. That disproportion is the whole point, and it is defensible only if it corresponds to something real about the work.

— 3. What a group weight does to a score

Illustrative figures throughout this section. Percentages are of the scored content within each group; the overall figure is a percentage of the total scored content. All results are rounded to one decimal place.

The weighting is realised **through the composition of the form, not through per-item score weights**. Every scored item counts the same. A form is assembled so that 40 per cent of its scored items come from Group 1, 40 per cent from Group 2 and 20 per cent from Group 3, and the score is then simply the proportion answered correctly. Those two descriptions are arithmetically identical — if the group item counts are $0.40N$, $0.40N$ and $0.20N$, then $(0.40N \cdot p_1 + 0.40N \cdot p_2 + 0.20N \cdot p_3) \div N = 0.40p_1 + 0.40p_2 + 0.20p_3$ — so the weighted formula below is the correct way to reason about a score, and no item is secretly worth more than another.

$$\text{Overall \%} = (0.40 \times \text{Group 1 \%}) + (0.40 \times \text{Group 2 \%}) + (0.20 \times \text{Group 3 \%})$$

3.1 Two candidates, one score

	Group 1 (40 %)	Group 2 (40 %)	Group 3 (20 %)	Overall
Candidate A — within-group score	80.0 %	70.0 %	40.0 %	
Candidate A — contribution	$0.40 \times 80.0 = 32.0$	$0.40 \times 70.0 = 28.0$	$0.20 \times 40.0 = 8.0$	68.0 %
Candidate B — within-group score	90.0 %	70.0 %	20.0 %	
Candidate B — contribution	$0.40 \times 90.0 = 36.0$	$0.40 \times 70.0 = 28.0$	$0.20 \times 20.0 = 4.0$	68.0 %

Both candidates score 68.0 per cent overall and both clear 65 per cent. Throughout this document, 65 per cent is used as the reference because it is the platform's **configured** pass mark; the definitive standard will be set by a modified-Angoff standard-setting study, and the arithmetic below works the same way against whatever standard that study sets. See [EXB-05 – Standard setting: modified Angoff and the pass mark](#).

The two candidates are not equivalent professionals. Candidate B is a stronger accountant and, on this evidence, has close to no governed-AI judgement at all — and the weighted total conceals that, which is exactly why the result carries a domain-level breakdown and why §6 discusses hurdles honestly rather than pretending the issue away.

3.2 What a weak group costs

A group you abandon does not cost you its weight; it costs you the whole of its weight, and raises the bar on everything else.

Score **zero** in the AI group and the maximum overall score available is $40 + 40 = \mathbf{80.0 \text{ per cent}}$. To reach 65 per cent from there, the average across the other two groups must be:

$$65 \div 0.80 = 81.25 \%$$

$$\text{Check: } (0.40 \times 81.25) + (0.40 \times 81.25) + (0.20 \times 0) = 32.5 + 32.5 + 0 = 65.0 \%$$

So a candidate who writes off Domain 13 has quietly agreed to score above 81 per cent across the other twelve domains — nineteen knowledge areas of accounting and thirty-five of project management principles — to pass. Very few candidates who plan to skip the AI content have done that arithmetic before they sit.

3.3 The marginal value of a point

Improving your within-group score by 20 percentage points is worth:

- **8.0 points overall** in either 40 per cent group ($0.40 \times 20 = 8.0$);
- **4.0 points overall** in the 20 per cent group ($0.20 \times 20 = 4.0$).

That is the correct way to prioritise revision when time is short: the weight tells you what a point of improvement is worth, and your own diagnostic tells you where a point is cheapest to buy. The two together, not either alone, give the plan.

3.4 The number nobody computes: implied weight per knowledge area

If coverage within a group were spread evenly across its knowledge areas — which the blueprint does not promise, and which §5 qualifies — the implied weight of a single knowledge area would be:

Group	Weight ÷ knowledge areas	Implied weight per knowledge area
Project accounting & finance	$40 \div 19$	2.11 %
Project management principles	$40 \div 35$	1.14 %
AI knowledge & practical approach	$20 \div 7$	2.86 %

Read the last column again. **A knowledge area in the AI group carries exactly two and a half times the implied weight of a knowledge area in project management principles** — $(20 \div 7) \div (40 \div 35) = 2.5$ — and about 1.36 times that of an accounting and finance knowledge area — $(20 \div 7) \div (40 \div 19) = 1.357$.

This is the single most useful figure in the document for a candidate deciding where the next study hour goes, and it inverts the intuition almost everyone starts with. Domain 13 looks like one domain out of thirteen. Per knowledge area, it is the densest content on the paper.

— 4. Why these three weights

4.1 Why 40 per cent for project accounting and finance

Because the failure mode this credential exists to prevent is a cost report that does not reconcile to the accounts, and the people who produce those reports are, as a population, weakest here.

A controls professional works at the boundary between the project's own measurement of itself and the entity's statutory position. Cost incurred, revenue recognised, work executed, amounts certified and amounts invoiced are five different numbers on the same contract in the same month.

A professional who cannot say why they differ, and which of them belongs in which report, cannot be relied on to report a project honestly — and will, under pressure, report whichever number is most comfortable.

The Institute's judgement is that this is the largest genuine competence gap in the practising population, and weighting is the only instrument an examination has for saying so. **PCI interpretation**, and stated as such.

4.2 Why 40 per cent for project management principles

Because it is the discipline itself, and because a wide surface examined thinly is still the surface a controls professional stands on.

This is the largest group by content — thirty-five knowledge areas covering cost control, earned value, commercial and contract management, the delivery lifecycle, adaptive delivery, scheduling, the business process cycles and risk. Weighted equally with a group half its size, each of its knowledge areas is sampled less densely. That is a real trade-off and it has a real cost: a candidate can be examined on proportionately less of what they do every day.

It is accepted deliberately. This is the content practitioners are most likely to have learned on the job, and an examination adds least where experience already teaches most. The value of the assessment is concentrated where practice does not reliably produce competence — which is §4.1 and §4.3.

4.3 Why 20 per cent for AI, and why that is not a gesture

Because a fifth of the paper is the smallest weight that cannot be revised away, and because the alternative positions are both worse.

A credential that weighted AI at five per cent would be making a claim about the discipline that the Institute does not believe: that governed use of these tools is a peripheral skill a professional can acquire later. A credential that weighted it at forty per cent would be certifying something other than project controls.

Twenty per cent has a specific property. It is large enough that a candidate cannot pass while ignoring the group — §3.2 shows what ignoring it actually requires — and small enough that the credential remains a project controls credential rather than a technology credential. Combined with the density figure in §3.4, it means Domain 13 must be studied properly rather than skimmed the week before.

The content is vendor-neutral and concerns technique, data, governance and accountability rather than products. That is what makes a 20 per cent weight defensible over a three-year certification cycle: tools change inside a cycle, and what the group examines does not.

4.4 AI is also examined outside its group

The 20 per cent is the group weight for Domain 13. It is not the total AI exposure on a form. Every domain in the body of knowledge carries embedded coverage of AI as it applies to that domain, and an item in Group 1 or Group 2 may put a machine-generated output in front of a candidate and ask what a competent professional does with it. Such items are scored to the domain of their subject matter, not to Domain 13 — see [EXB-01](#) §6. A candidate estimating their AI exposure from the group weight alone is estimating low.

— 5. What the weighting does not do

It sets no per-domain weight. Weights attach to the three groups. The Institute publishes no percentage for any individual domain and will not confirm one informally. If coverage within a group were spread evenly across its member domains, Group 1 would imply about 10 per cent per domain ($40 \div 4$), Group 2 about 5 per cent ($40 \div 8$) and Group 3 the full 20 per cent — but the blueprint does not promise even spread, and those figures are an illustration of the structure, not a published specification. Treat them as a sanity check on your revision plan and nothing more.

It sets no per-domain hurdle. The domain breakdown on a result is diagnostic. There is no requirement to reach a threshold in each domain separately.

It does not fix how many items each group contains. The weighting is a proportion, and a proportion of an unknown total is still unknown in absolute terms. The number of scored items in each group will follow from the total, which **will be confirmed by a formal job-task analysis** — **[CONFIRM: number of scored items per group, once the job-task analysis fixes the total]**. See [EXB-06 – Job-task analysis: where the blueprint gets its authority](#).

It is not itself the passing standard. A weighting decides what the score is made of. The standard decides how much of it is enough. Those are separate decisions taken by separate methods; the second is in [EXB-05 – Standard setting: modified Angoff and the pass mark](#).

— 6. The honest limitation: a compensatory model has a floor problem

This is the objection a competent assessment reviewer raises first, and it deserves the arithmetic rather than a deflection.

The model is **compensatory**: strength in one group offsets weakness in another. Follow that to its boundary. A candidate scoring 100 per cent in project accounting and finance and 100 per cent in the AI group has already banked $40 + 20 = 60$ points. They need five more, and five points from a 40 per cent group requires:

$$5 \div 0.40 = 12.5 \%$$

$$\text{Check: } (0.40 \times 100) + (0.40 \times 12.5) + (0.20 \times 100) = 40.0 + 5.0 + 20.0 = 65.0 \%$$

So on the arithmetic alone, a candidate could score 12.5 per cent across the whole of project management principles — cost control, earned value, contracts, lifecycle, scheduling, process cycles and risk — and still pass. By symmetry the same is true of the accounting and finance group.

Two things should be said about that, and the Institute declines to say only the comforting one.

The scenario is close to unreachable in practice. Two perfect group scores are extraordinarily rare, and the domain-crossing design of the items means a candidate who genuinely cannot read a schedule or an earned value position will lose marks inside the other groups as well, because the scenarios there assume that competence. The arithmetic boundary and the empirical boundary are not the same place.

But "unlikely" is not a control. Whether the PCL-AI should carry a per-group hurdle — a minimum within-group score required in addition to the overall standard — is a genuine question on which competent assessment professionals differ. A hurdle protects against exactly the profile above. It also raises the overall difficulty in a way that is hard to justify without evidence, intro-

duces a second decision point with its own measurement error, and can fail candidates who are competent overall on the basis of a small number of items in one group. **Professional judgement**, and labelled as such.

The Institute's position today: **no hurdle is published, and none is applied**. The question will be put to the job-task analysis and the standard-setting study, which are the only instruments that can answer it with evidence rather than argument. If a hurdle is introduced, it will be published as a major version of [EXB-01](#) with an effective date and a notice period, and it will not be applied retrospectively.

— 7. How this goes wrong

Treating the group weight as a difficulty rating. A 20 per cent weight does not mean the AI items are easier, shorter or more forgiving. It means there are fewer of them. Weight and difficulty are unrelated quantities, and candidates routinely conflate them.

Revising by domain count instead of by weight. Thirteen domains invites thirteen equal blocks of study time. That allocation gives Domain 13 about 7.7 per cent of the effort against 20 per cent of the paper, and over-serves Group 2 by a wide margin. The arithmetic in §3.4 is the correction.

Assuming the weighting maps onto the day job. It does not, and it is not meant to. A cost engineer may spend ninety per cent of the week inside Group 2 content. The examination weights what the credential attests, not what a particular role happens to do this year.

Reading a strong overall score as uniform competence. Candidate B in §3.1 passed comfortably with close to nothing in the AI group. That is what a compensatory model permits. Employers reading a PCL-AI result should read the domain breakdown, not only the outcome.

Quoting a per-domain percentage. Some legacy material carries indicative per-domain figures from an earlier grouping of the same thirteen domains. Those figures are superseded and are not the published specification. The current published weighting is the three group weights in §2 and nothing else.

— 8. Checklist

For a candidate planning revision:

- [] I have converted my diagnostic scores into weighted contributions using the formula in §3, not compared raw domain percentages.
- [] I know what my current profile scores overall, and by how much it clears or misses 65 per cent.
- [] I have computed what score I would need elsewhere if I left my weakest group where it is.
- [] My study hours are allocated against the implied per-knowledge-area weights in §3.4, not evenly across thirteen domains.
- [] I have not assumed that a lower group weight means easier items.

For a training provider or reviewer:

- [] Course hours are allocated against the published group weights, and the allocation is stated to learners.

- [] The course teaches the density point in §3.4 explicitly, rather than letting learners infer effort from domain count.
 - [] No per-domain percentage is quoted to learners, because none is published.
 - [] The compensatory model and its floor problem in §6 are disclosed rather than glossed.
-

A weighting is the one part of an examination specification that changes candidate behaviour before anyone sits it. Publishing ours with the arithmetic attached means a candidate can check our reasoning, a provider can build against it, and a reviewer can hold us to it when the job-task analysis reports — which is the point at which this weighting stops being a considered judgement and starts being an evidenced one, or gets changed.

— Related

- [EXB-01 – Examination blueprint – PCL-AI](#) — the authoritative statement of the weighting, the syllabus and the format
- [EXB-05 – Standard setting: modified Angoff and the pass mark](#) — how much of the weighted score is enough, and who decides
- [EXB-06 – Job-task analysis: where the blueprint gets its authority](#) — the study that will test this weighting against evidence
- [PCB-03 – Why 40/40/20 – the weighting and what it claims](#) — the same decision read from the body of knowledge rather than the examination

— Sources and standards

The weighting, domain structure and knowledge-area counts are taken from the PCI Body of Knowledge ([docs/bok/](#)) and the published blueprint in [EXB-01](#). All arithmetic in §3 and §6 is illustrative, computed from those published proportions, and independently recomputed before submission.

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EXB-03 · EXAM BLUEPRINT

Cognitive levels and item types

What Recall, Application and Analysis mean here, why the examination uses four-option single-best-answer items, and what that format cannot measure

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

Every PCL-AI item is tagged at one of three cognitive levels — Recall, Application or Analysis — and the intended mix is application-weighted. This document defines each level with the kind of stem it produces, explains the anatomy of a four-option single-best-answer item, gives the Institute's reasons for choosing that format over the alternatives, and sets out honestly what the format cannot measure and what is done about each limitation. It also lists the item types the Institute refuses to use, and why. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

— About this document

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Cognitive levels and item types

In one paragraph. Every PCL-AI item is tagged at one of three cognitive levels — Recall, Application or Analysis — and the intended mix is application-weighted. This document defines each level with the kind of stem it produces, explains the anatomy of a four-option single-best-answer item, gives the Institute's reasons for choosing that format over the alternatives, and sets out honestly what the format cannot measure and what is done about each limitation. It also lists the item types the Institute refuses to use, and why. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

Who this is for. Candidates who want to know what kind of thinking the examination rewards; item writers and reviewers working to the Institute's specification; and assessment professionals or academics auditing the design.

— 1. The level matters more than the topic

Two items can sample the same knowledge area and measure entirely different things.

Which formula gives the estimate at completion when past cost performance is expected to continue?
Cost inefficiency on this project is systemic and the schedule driver has been removed. Which estimate at completion is the most defensible forecast, and why?

Both sit in knowledge area 6.3. The first asks whether a candidate has memorised a relationship. The second asks whether they can read a situation, choose between defensible methods, and justify the choice. A credential built mainly from the first kind certifies revision. A credential built mainly from the second kind certifies judgement.

This is why every item carries a cognitive level as well as a domain and knowledge-area tag, and why form assembly is constrained on both. Content sampling alone — the subject of [EXB-01 – Examination blueprint – PCL-AI §5](#) — guarantees that the right topics appear. Only the level constraint guarantees that they appear at the right depth.

— 2. The three levels

2.1 Recall

What it asks. State or recognise a definition, a rule, a relationship or what a standard requires.

What it proves. That the candidate holds precise knowledge that decisions depend on. Recall is not a lesser skill; it is a prerequisite one. A professional who thinks contingency reserve and management reserve are the same thing will misreport a baseline change as a variance, and no amount of analytical ability repairs that.

Typical stem shapes. Which statement about X is correct? · Under this standard, revenue is recognised when... · The performance measurement baseline is revised...

Where the Institute uses it. Where precision genuinely changes a decision: definitional boundaries that practitioners routinely blur, recognition tests, the distinction between two adjacent concepts. Not as filler, and not for vocabulary that a professional would reasonably look up.

2.2 Application

What it asks. Use a known method on given data — compute a figure, classify a transaction, apply a rule to a stated situation.

What it proves. That the candidate can convert knowledge into a number or a classification that someone else could rely on. This is the level at which most controls work actually happens, and it is where a candidate's understanding either survives contact with data or does not.

Typical stem shapes. Given these figures, what is...? · At cut-off, what entry does the cost report require? · Applying the published threshold, which control accounts are flagged?

Where the Institute uses it. Throughout, and especially in the quantitative knowledge areas — earned value and its indices, the estimate-at-completion family, variance decomposition, progress valuation and payment, expected value, agile forecasting. Application items supply the data and never the method; the formula policy in [EXB-01](#) §7.2 exists precisely so that selecting the method remains part of the assessment.

2.3 Analysis

What it asks. Interpret a situation in which more than one course of action is arguable, weigh the competing considerations, and select what a competent professional would defensibly do next.

What it proves. Professional judgement under incomplete information — which is what a Leader-level credential is for. An Analysis item usually contains no arithmetic error to spot. It contains a position, and the question is what you do about it.

Typical stem shapes. Given this position, what should the controls professional do next? · The most accurate statement of the position is... · The most defensible response is...

Where the Institute uses it. Wherever the discipline's real difficulty lies: choosing between two legitimate forecasting methods on the evidence, deciding whether an instructed variation is a cost or a trend, judging whether a machine-generated output can be relied on, deciding what to escalate and when.

What an Analysis item never rewards. The heroic answer. The best option is what a practitioner could defensibly do next — often verifying a figure, correcting a baseline, or telling someone something they do not want to hear — not a dramatic intervention that no controls professional has the authority to make.

— 3. The intended mix

The examination is **application-weighted**. The majority of scored items sit at Application and Analysis; Recall is the minority of the paper and appears where precise knowledge genuinely matters.

That is a statement about design intent, and it is as far as the Institute will go today. The exact proportions form part of the examination specification and will be confirmed through item development and the job-task analysis: **[CONFIRM: target proportion of items at each cognitive level, to be fixed in the examination specification]**. Publishing a precise mix before the item bank can support it would be a number invented to look rigorous.

The honest statement a candidate can act on: **expect to be tested mainly on doing and judging**. If your preparation consists of memorising definitions and formulae, you have prepared for the smallest part of the paper.

— 4. The item format

4.1 Anatomy

Every item is a **four-option multiple-choice question with a single best answer**. It has three parts.

The stem sets out the situation and supplies every figure needed to answer it. In a scenario item the stem is drawn from real controls work — a cost position at month end, a network with a delay, a payment application, a machine-generated forecast — and contains no information the candidate is expected to supply from outside the body of knowledge.

The key is the single best answer. Not merely a correct answer: the best of the four, judged against the standard the body of knowledge teaches. Where two options are both technically permissible, the key is the one a competent professional would choose on the evidence given, and the rationale says why the other is second best rather than wrong.

The three distractors are wrong, and wrong for a reason. Every distractor in a well-written item corresponds to a specific error a real practitioner makes: an inverted ratio, a cumulative figure used where an incremental one is required, a method applied outside its assumptions, a plausible action that contradicts something stated in the stem. A distractor nobody would choose measures nothing and wastes a quarter of the item.

4.2 Scoring mechanics a candidate should know

- **Every scored item carries equal weight.** The 40/40/20 weighting is achieved through the composition of the form, not through per-item weights — see [EXB-02 – Domain weightings and the 40/40/20 rule](#) §3.
- **There is no negative marking.** A wrong answer scores zero. It does not score less than zero.
- **An unanswered item scores exactly what a wrong answer scores.** There is therefore never a reason to leave an item blank. If you are out of time, answer.
- **Random guessing on a four-option item has an expected value of 25 per cent ($1 \div 4$).** That is the chance floor, and it is well below any plausible passing standard — which is one reason the standard has to be set by a documented method rather than chosen as a round number. See [EXB-05 – Standard setting: modified Angoff and the pass mark](#).

4.3 How to work an item

The discipline that answers items well is the discipline the examination is trying to measure.

1. **Establish the position before reading the options.** State in your own words what the data actually says — what the indices mean, what the dates imply, what the ledger is and is not capturing. Options read first will frame the data; data read first will filter the options.
2. **Eliminate what contradicts the stem.** The most common weak distractor is a reasonable-sounding action that ignores something the scenario has already told you.
3. **Choose the defensible action, not the impressive one.**
4. **Apply the same test to any machine output in the stem.** Ask whether it is explainable, validated and owned before anything is built on it. If the option builds on an unvalidated output, it is not the key.

— 5. Why this format

The choice of a single item type across an entire examination needs a defence, because it is also a constraint.

Scoring is objective and reproducible. A key is a key. Two markers cannot disagree, a candidate cannot be advantaged by an examiner's mood or handwriting, and a challenge to a result is answered by inspecting the item rather than by re-marking a judgement. For a certification decision that must be defensible to the candidate, the employer and any appeal, this matters more than it does in education.

It is auditable at the item level. Because every response is a discrete choice, each item generates statistics — how many candidates got it right, and whether the ones who got it right were the stronger candidates overall. That is what makes item quality measurable rather than assertable, and it is the basis of the whole review and retirement process in [EXB-04 – How items are written, reviewed and retired](#).

It samples broadly in the time available. Ninety minutes of constructed response would cover a handful of knowledge areas. The blueprint commits the examination to sampling across sixty-one, and only a compact item type makes that arithmetic work. Breadth of sampling is itself a fairness property: the narrower the sample, the more a result depends on which topics happened to appear.

It delivers and translates reliably. The examination interface is available in seven languages. A discrete-choice item can be translated and reviewed for equivalence in a way that a marked essay, judged against a rubric applied by markers working in different languages, cannot.

Scenario design recovers most of what the format appears to lose. The common objection is that multiple-choice measures recall. That is true of badly written multiple-choice. An item that presents a project position and asks what to do next is measuring judgement; the response format is just how the judgement is captured.

— 6. What this format cannot do

Every one of these is a real limitation. The Institute states them rather than waiting to be told.

It measures recognition, not generation. A candidate who can identify the correct estimate at completion among four has not demonstrated that they could build one from a blank sheet. Recognition and generation are related but not identical abilities, and the format only reaches the first.

What we do about it. Scenario stems supply raw data rather than pre-computed intermediate figures, so the candidate must generate the intermediate result before any option becomes recognisable. An item that gives you the cost performance index has tested less than one that gives you earned value and actual cost.

Options cue. Seeing four possibilities constrains and prompts thinking in a way an open question does not. Distractor sets can also leak the answer through length, specificity or grammatical fit.

What we do about it. Item-writing rules require options of comparable length and grammatical structure, prohibit the key being systematically the longest or most qualified option, and require reviewers to attempt the item with the stem covered. An item answerable from the options alone is rejected.

There is no partial credit and no visible working. A candidate who selects the right method and makes an arithmetic slip scores the same as one who selected the wrong method entirely. The format sees the answer, never the reasoning.

What we do about it. We accept it, and design around it. Distractors are built from the specific intermediate errors a candidate would make, so the pattern of a candidate's wrong answers is informative even though a single item is not. This is also why the domain breakdown on a result matters: it is the only diagnostic the format can offer.

Guessing contributes. At four options, chance alone returns 25 per cent.

What we do about it. The standard-setting method in [EXB-05](#) addresses chance directly: a judge asked what proportion of minimally competent candidates would answer an item correctly is implicitly rating against the chance floor, and an item that panellists rate near or below 25 per cent is flagged as flawed rather than merely hard.

It cannot assess a sustained piece of professional work. Building a cost breakdown structure, negotiating a variation, writing a defensible basis of estimate, presenting a bad forecast to a sponsor — none of these is reachable by any multiple-choice item, however well written.

What we do about it. Nothing, within this examination — and the Institute says so rather than implying otherwise. A PCL-AI attests assessed knowledge, applied method and professional judgement at the point of assessment. It does not attest observed workplace performance, and no examination of this type does. The competency framework in [CMP-03 – PCL-AI: the fourteen competencies](#) describes what evidence of the wider capability looks like; the examination is one input to a certification decision, not the whole of it.

— 7. Item types the Institute does not use

Each of these appears in professional examinations, and each is excluded deliberately.

Excluded type	Why
True/false	A 50 per cent chance floor, and almost no capacity to carry a scenario.
Multiple response ("select all that apply")	Scoring is contested — partial credit, all-or-nothing and negative marking all give different results from the same response — and none of the schemes is obviously fairest. The complexity buys no measurement.
Negatively worded stems ("which is NOT correct?")	They test careful reading of the question rather than knowledge of the answer, and they translate badly across seven interface languages.
"All of the above" / "None of the above"	Partial knowledge of two options resolves the item without reading the third. They also make distractor analysis meaningless.
Complex multiple choice (combinations such as "i and iii only")	Cueing is severe, difficulty comes from the combinatorics rather than the content, and the statistics are hard to interpret.
Items turning on the wording of a protected standard	Reproducing protected text is prohibited, and a candidate should be assessed on what a requirement means, not on whether they have memorised its phrasing.

— 8. How this goes wrong

Preparing at the wrong level. The commonest and most expensive candidate error. Flashcards and formula memorisation prepare for the smallest part of an application-weighted paper. Working full scenarios under time pressure prepares for the largest.

Treating "single best answer" as "only correct answer". In an Analysis item, more than one option may be defensible in some circumstance. The task is to choose the best on the evidence given. A candidate who argues with the item instead of answering it loses the mark.

Reading the options first. It feels faster and reliably produces worse answers, because the option set frames the data.

Leaving items blank. There is no negative marking. A blank is a guaranteed zero where a guess is a one-in-four chance.

Item writers letting the key grow. The most persistent defect in submitted items is a key that is longer and more qualified than the distractors, because the writer wanted to make it unambiguously right. Length is a cue, and reviewers reject on it. The worked commentary in [EXB-08 – Sample items with rationale](#) shows the correction.

Mistaking a hard item for a good one. Difficulty and quality are independent. An item is good when the candidates who answer it correctly are the candidates who are stronger overall; that is a measurable property, and it is the subject of [EXB-04](#).

— 9. Checklist

For a candidate:

- [] I have practised at least one full timed set rather than only individual items.
- [] For quantitative items, I compute the intermediate figure before looking at the options.
- [] I can state, for any item I get wrong, which specific error the distractor I chose corresponds to.

- [] I never leave an item unanswered.
- [] I can recognise an Analysis item and resist the impressive option.

For an item writer or reviewer:

- [] The item carries a domain, knowledge-area and cognitive-level tag, and the level matches what the item actually demands.
- [] The stem supplies every figure needed and no figure that is not needed, except where a redundant figure is a deliberate and explained test.
- [] The item is unanswerable from the options alone — verified by attempting it with the stem covered.
- [] Options are of comparable length and parallel grammatical structure; the key is not the longest.
- [] Every distractor maps to a named, real practitioner error, recorded in the rationale.
- [] The item uses none of the excluded types in §7.
- [] No protected standard text is reproduced anywhere in the item.

The reason to publish the limits of a format alongside its defence is that a candidate, an employer and an auditor are each entitled to know what a result does and does not evidence. A PCL-AI pass says that, on the day, against a published sampling frame, the holder could read a project position and decide what to do about it. It does not say they have been observed doing the job. Any certification body that lets those two claims blur has stopped measuring and started marketing.

— Related

- [EXB-01 – Examination blueprint – PCL-AI](#) — the sampling frame, the format specification and the formula policy
- [EXB-04 – How items are written, reviewed and retired](#) — the process that turns these rules into a bank
- [EXB-05 – Standard setting: modified Angoff and the pass mark](#) — how the chance floor and item difficulty enter the passing standard
- [EXB-08 – Sample items with rationale](#) — the format worked in full, with every distractor explained

— Sources and standards

The cognitive-level definitions are the Institute's own, expressed in its own words and aligned to the levels used to tag the published sample material in [docs/bok/](#). The item-type exclusions in §7 reflect long-standing item-writing practice in professional certification; they are stated here as the Institute's rules rather than attributed to any single published source.

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Status and version

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EXB-04 · EXAM BLUEPRINT

How items are written, reviewed and retired

The item lifecycle the Institute is standing up — commissioning, authoring, review, bias review, pretesting, statistics and retirement

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

An examination is only as good as its item bank, and an item bank is a managed asset with a lifecycle rather than a folder of questions. This document sets out the lifecycle the Institute is standing up for the PCL-AI: how items are commissioned against the blueprint, the rules they are authored to, the three independent reviews they must survive, how they are pretested unscored, the statistics used to judge them once real candidates have answered them, and the triggers that retire them. It works the statistics through with illustrative figures and states plainly which parts of the process are running today and which are being built. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

— About this document

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How items are written, reviewed and retired

In one paragraph. An examination is only as good as its item bank, and an item bank is a managed asset with a lifecycle rather than a folder of questions. This document sets out the lifecycle the Institute is standing up for the PCL-AI: how items are commissioned against the blueprint, the rules they are authored to, the three independent reviews they must survive, how they are pre-tested unscored, the statistics used to judge them once real candidates have answered them, and the triggers that retire them. It works the statistics through with illustrative figures and states plainly which parts of the process are running today and which are being built. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

Who this is for. Subject-matter experts writing or reviewing items for the Institute; assessment professionals auditing our process; and practitioners who want to know why an examination body's quality claims should be believed.

— 1. The bank is the asset

The blueprint says what should be sampled. The item bank is what makes the sampling possible, and it is the single most expensive and most fragile thing a certification body owns.

Fragile, because an item bank degrades in four directions at once. Items leak, and a leaked item measures nothing. Practice changes, and an item that was correct in 2026 becomes wrong or merely quaint. The bank gets used, and over-exposed items stop discriminating. And every item carries a small probability of being defective in a way that no amount of expert authorship prevents and only candidate data reveals.

A lifecycle is the answer to all four. What follows is the process the Institute is building for the PCL-AI bank. §11 says honestly which parts of it are operating now.

— 2. The lifecycle in one view

Stage	Who	What happens	What stops an item here
1. Commission	Chief Examiner / Assessment Lead	A gap in the bank is identified against the blueprint and an item is commissioned to a specific knowledge area and cognitive level	—
2. Author	Subject-matter expert	The item is written to the authoring rules in §4, with a rationale for the key and for every distractor	—
3. Technical review	A second subject-matter expert	Correctness of the stem, the key, the arithmetic and every distractor rationale	Wrong key; unverifiable arithmetic; two defensible answers
4. Editorial review	Editorial	Clarity, format compliance, option balance, language, translation readiness	Cueing; excluded item types; protected text
5. Bias and fairness review	A panel independent of the author	Construct-irrelevant barriers: culture, jurisdiction, sector, language, accessibility	Anything that makes the item harder for reasons unrelated to competence
6. Pretest	—	The item is placed on live forms unscored , and candidate responses are collected	—
7. Statistical review	Assessment Lead with psychometric advice	Difficulty, discrimination and distractor performance are computed and read	Statistics outside the acceptance thresholds
8. Disposition	Chief Examiner	Accept into the scored bank, return for revision, or reject	—
9. In-bank management	Assessment Lead	Exposure tracking, periodic content review, version control	—
10. Retirement	Chief Examiner	The item leaves the scored bank on one of the triggers in §10	—

No item reaches a candidate's score without passing every stage. An item that fails a stage returns to its author with the reason; it is not quietly fixed by the reviewer, because a reviewer who edits an item has just reviewed their own work.

— 3. Commissioning: writing to a gap, not to a favourite topic

Items are commissioned against the blueprint, not offered against it. The Assessment Lead identifies where the bank is thin — a knowledge area with too few usable items, a cognitive level under-supplied in a particular group, a domain whose items are ageing — and commissions to that gap with the domain, knowledge area, cognitive level and, where relevant, the scenario setting specified in advance.

The alternative, which is how most banks accumulate, is to let experts submit what interests them. It produces a bank that is deep where the authors are strong and thin everywhere else, and it makes the blueprint aspirational: forms cannot be assembled to a specification the bank cannot supply.

Commissioning also spreads sector settings deliberately. A bank in which every scenario is a construction project examines construction professionals more gently than everyone else, and that is a fairness defect before it is an editorial one.

— 4. Authoring: the rules

Authors work to the format specification in [EXB-03 – Cognitive levels and item types](#) and to these rules.

One assessable point per item. If an item can be got wrong for two unrelated reasons, its statistics will be uninterpretable and its diagnostic value nil.

The stem stands alone. A competent candidate should be able to answer before seeing the options. Every figure needed is in the stem; nothing needed is outside the body of knowledge.

Four plausible options, of comparable length and parallel structure. The key is never systematically the longest, the most qualified or the most technically phrased. This is the most common defect in submitted items and the easiest for a candidate to exploit.

Every distractor names a real error. The author records, for each distractor, the specific mistake a practitioner would make to arrive at it — an inverted ratio, a cumulative figure used where an incremental one is needed, a method applied outside its assumptions, an action that contradicts the stem. A distractor without a named error is a filler option and is returned.

Arithmetic is shown in the rationale, not just the answer. $CPI = EV \div AC = 4,500,000 \div 5,000,000 = 0.90$. The reviewer recomputes from the substitution.

Currency-neutral, jurisdiction-neutral, vendor-neutral. Where a legal, tax or accounting treatment varies, the item states the frame as an assumption or avoids depending on it. Where a tool is involved, the item concerns the technique, never a product's interface.

Nothing protected is reproduced. Standards are described in the Institute's own words. No text, table, diagram or question from any external standard or examination is reproduced in any item.

No live content is ever published. Items written for publication — the sample items in [EXB-08 – Sample items with rationale](#) — are authored separately, marked as study material, and never enter the live bank. The separation runs in both directions: a live item is never published, and a published item is never scored.

— 5. Technical review: is it right?

A second subject-matter expert, who did not write the item, verifies four things.

The key is correct, and correct for the reason the author gives.

Every distractor is wrong, and wrong for the reason the author gives. This is where items most often fail. An author writes what they believe is a clearly inferior option; a reviewer from a different sector recognises it as standard practice in theirs. An item with two defensible answers is not a hard item, it is a broken one.

Every calculation is independently recomputed from the substitution, not checked against the author's answer. Numerical items get two technical reviewers.

The cognitive level tag matches what the item demands. An item tagged Analysis that resolves to a formula lookup is re-tagged or rewritten. Mis-tagged items corrupt form assembly silently, because the assembly rules trust the tag.

— 6. Bias and fairness review

This is a distinct review with a distinct question, and it is not a politeness check.

The question is: **does anything in this item make it harder for a group of candidates for reasons that have nothing to do with the competence being assessed?** Assessment calls that construct-irrelevant difficulty, and it is the mechanism by which a technically flawless item can still be unfair.

The panel reads every item for:

Jurisdictional assumption. An item that turns on one country's tax treatment, procurement law, retention practice or accounting rule tests where the candidate works, not what they know. The Institute's content is globally applicable; where treatment varies, the item states the frame as an assumption of the scenario.

Sector assumption. A scenario that requires knowledge of how a specific industry operates — commissioning sequences, shift patterns, regulatory gates — advantages that industry's practitioners. Scenarios carry the context they need inside the stem.

Cultural and idiomatic content. Idiom, humour, sporting or culinary metaphor, names and settings that read as familiar to some candidates and as noise to others. The examination interface runs in seven languages; an item that survives translation only as a paraphrase is a defective item.

Language complexity beyond the construct. Long subordinate clauses, rare vocabulary and unnecessary negation measure reading comprehension. Where a term is genuinely part of the discipline, it stays; where it is decoration, it goes.

Accessibility. No meaning carried by colour alone. No dependence on a visual layout that a screen reader cannot linearise. Figures and tables that read sensibly in sequence.

The panel is independent of the author, and it includes reviewers from more than one region and more than one delivery sector. A review panel drawn from a single background cannot see its own assumptions — which is the whole reason the stage exists.

Bias review before pretesting catches what can be reasoned about. It does not catch everything, which is why §7.3 exists.

— 7. Pretesting and item statistics

7.1 Why items are pretested unscored

A new item is placed on live forms as an **unscored** item. Candidates cannot tell which items are scored and which are being trialled, and pretest items do not contribute to anyone's result.

This is the only honest way to get the data. Item statistics computed on a volunteer sample of trainees describe trainees; statistics computed on real candidates under real examination conditions de-

scribe the population the item will actually be used on. The unscored placement means no candidate is ever penalised for an item the Institute has not yet validated.

An item is not scored until it has accumulated enough responses to make its statistics meaningful: [CONFIRM: minimum pretest response count before an item may be scored].

7.2 The two statistics that matter

Difficulty is the proportion of candidates who answered correctly — conventionally the p-value. Confusingly for newcomers, a high p-value means an easy item. On a four-option item, chance alone returns 0.25, so a p-value near or below 0.25 means candidates are not doing better than guessing.

Discrimination is whether the candidates who answered the item correctly were the candidates who were stronger overall. It is the more important of the two, because it is the property that makes an item useful: an item everyone gets right and an item everyone gets wrong both discriminate nothing, whatever their difficulty. The Institute computes a **discrimination index** — the proportion correct in the strongest group minus the proportion correct in the weakest group, using the conventional upper and lower 27 per cent of candidates by total score — alongside the item-total correlation.

A **negative discrimination index is the strongest signal in the whole process**. It means stronger candidates did worse on the item than weaker candidates, which almost always means the key is wrong, a distractor is also defensible, or the stem is misleading in a way that only expertise exposes.

7.3 Distractor analysis

The option counts are read as well as the totals. Three patterns get an item looked at:

- **A distractor drawing more responses than the key** — investigate immediately for a mis-key or a second defensible answer.
- **An option almost nobody selects** — a dead distractor. The item is effectively three-option, and its chance floor rises from 25 per cent to 33.3 per cent ($1 \div 3$), which changes what its difficulty means.
- **A distractor drawing strong candidates disproportionately** — usually an option that is correct under a reading the author did not anticipate.

Distractor analysis is also where bias review gets its second bite: an item whose statistics differ between candidate groups after controlling for overall ability is flagged for differential item functioning review. That analysis identifies that an item behaves differently; it never explains why, so a flagged item goes back to a human panel rather than being deleted on the strength of a number.

7.4 Worked example

Illustrative figures. Response counts are for a single pretest item; percentages are rounded to two decimal places and the discrimination index to two.

Item A — accepted. 200 candidates responded. Key is B.

Option	Responses	Share
A	24	12.0 %
B (key)	122	61.0 %
C	38	19.0 %
D	16	8.0 %

Difficulty: $p = 122 \div 200 = 0.61$ — moderately difficult, comfortably above the 0.25 chance floor.

Discrimination, using the upper and lower 27 per cent ($0.27 \times 200 = 54$ candidates in each group). Of the 54 strongest candidates, 45 answered correctly; of the 54 weakest, 14 did.

$$D = (45 \div 54) - (14 \div 54) = 0.8333 - 0.2593 = 0.57$$

A strongly discriminating item. Every distractor is alive — the smallest draws 8 per cent — so all three are doing work. **Disposition: accept into the scored bank.**

Item B — rejected. 200 candidates responded. Key is C.

Option	Responses	Share
A	61	30.5 %
B	58	29.0 %
C (key)	55	27.5 %
D	26	13.0 %

Difficulty: $p = 55 \div 200 = 0.275$ — barely above the 0.25 chance floor, and a distractor has out-drawn the key.

Discrimination: of the 54 strongest candidates, 16 answered correctly; of the 54 weakest, 18 did.

$$D = (16 \div 54) - (18 \div 54) = 0.2963 - 0.3333 = -0.04$$

Negative. Stronger candidates did slightly worse than weaker ones, and the responses are spread across three options almost evenly — the signature of candidates guessing between options they cannot separate. **Disposition: reject.** The item goes back with the finding that option A is very likely defensible under a reading the author did not intend; it is rewritten from the stem up, not patched.

The formal acceptance thresholds for difficulty and discrimination — the values at which an item is accepted, flagged for review or rejected — are [\[CONFIRM: item-acceptance thresholds for difficulty and discrimination\]](#), to be fixed with psychometric advice and published. The Institute would rather name that gap than publish a threshold chosen because it looks like the sort of number an examination body ought to have.

— 8. Disposition

Three outcomes, decided by the Chief Examiner on the evidence from stages 5 to 7:

Accept — the item enters the scored bank with its tags, its rationale and its statistics attached.

Revise — the item returns to its author with the specific finding. A revised item re-enters at stage 3 and is pretested again. Revision does not preserve the old statistics: an item whose stem or options have changed is a new item, and treating it otherwise is how contaminated data enters a bank.

Reject — the item is retained with its findings for the author's learning and never used. Rejected items are kept, not deleted, because a bank's rejection record is the evidence that its review actually bites.

— 9. In-bank management

An accepted item is not finished.

Exposure is tracked. Every use of an item on a form is recorded. An item used too often becomes predictable, and over-exposure is a security matter as much as a psychometric one — see [EXB-07 – Examination security and integrity](#).

Statistics are monitored continuously. An item's difficulty drifting downwards over time — more candidates getting it right without any change to the population — is one of the clearest indicators that the item has leaked.

Content is reviewed periodically. Every item carries a review date. Domain 13 items age fastest, because the practice they describe moves fastest; items resting on a standard that has been amended are reviewed on the amendment, not on the calendar.

Versions are controlled. Every item version carries an identifier, and every scored attempt records the bank version and answer-key version it was scored against. A result can therefore always be reconstructed exactly as it was computed, which is what makes an appeal answerable years later.

— 10. Retirement

An item leaves the scored bank on any of these triggers:

Trigger	Why
Suspected or confirmed exposure	A leaked item measures familiarity with the leak. Retirement is immediate and does not wait for proof.
Over-exposure	Use beyond the tracked threshold, whether or not a leak is suspected.
Statistical drift	Difficulty or discrimination moving outside the acceptance range on continuing use.
Content superseded	A standard amended, a practice changed, a technique obsolete. Common in Domain 13.
Blueprint change	A knowledge area removed or restructured in a major version of EXB-01 .
Fairness finding	A differential-item-functioning flag confirmed by panel review.

Retirement is permanent for that item version. Retired items are archived with their full history — statistics, review findings and retirement reason — because the archive is what allows a future re-

viewer to ask whether the bank's failures cluster anywhere, which is a question no live bank can answer about itself.

Results already issued are **not** disturbed by retirement. A candidate scored against the bank version in force at their sitting keeps that result; retiring an item changes future forms, not past ones. The single exception is a confirmed mis-key, where the affected results are rescored and every affected candidate is notified — including those whose result improves and those for whom nothing changes.

— 11. What is running today, and what is being built

The Institute is founding-stage, and this section says which is which rather than describing an aspiration in the present tense.

Operating now. Commissioning against the published blueprint. Authoring rules and the rationale requirement. Technical review by a second subject-matter expert with independent recomputation of every calculation. Editorial review. Item versioning, with the bank version and answer-key version recorded against every scored attempt, so any result can be reconstructed exactly.

Being stood up. The independent bias and fairness panel as a standing body with defined regional and sector composition. Systematic unscored pretesting on live forms. Routine computation and review of item statistics, and the acceptance thresholds those reviews will apply. Differential-item-functioning analysis, which needs candidate volumes the Institute does not yet have.

Dependent on volume. Several of the statistics above are meaningless below a certain number of responses, and no process design can create that data. Until the volumes exist, the Institute relies more heavily on the human review stages and says so, rather than reporting statistics that a small sample cannot support.

— 12. How this goes wrong

Experts assuming expertise is enough. The commonest failure in professional certification. A distinguished practitioner writes an item that is correct, important and unanswerable — or answerable from the options alone. Authorship and item writing are different skills, and the second is learned.

Reviewing your own item. A reviewer who edits rather than returns has removed the independence the stage exists to provide. The rule is absolute: findings go back to the author.

Patching a failed item instead of rewriting it. Changing one distractor on an item with negative discrimination usually leaves the underlying ambiguity in place, and the patched item carries the old statistics as though they still applied.

Treating difficulty as quality. A hard item that everyone answers by guessing is worse than an easy item that separates strong candidates from weak ones. Discrimination is the property that matters.

Letting the bank age quietly. No trigger fires when an item becomes out of date; someone has to look. The review date on every item is the mechanism, and it only works if it is enforced.

Publishing sample items drawn from the live bank. The single most damaging thing a certification body can do to its own examination, and it is almost always done with good intentions — a training partner asks for realistic practice material. The separation in §4 is not negotiable.

— 13. Checklist

For an item author, before submission:

- [] Domain, knowledge area and cognitive level are tagged, and the level matches what the item demands.
- [] The item can be answered from the stem alone.
- [] Options are of comparable length and parallel structure; the key is not the longest.
- [] Every distractor has a named practitioner error recorded in the rationale.
- [] Every calculation shows the substitution, and I have recomputed it from scratch.
- [] The item is jurisdiction-neutral, sector-neutral and vendor-neutral, or states its frame as an assumption.
- [] Nothing protected is reproduced.
- [] None of the excluded item types in [EXB-03 §7](#) is used.

For a reviewer:

- [] I did not write this item.
 - [] I attempted it with the options covered before reading them.
 - [] I recomputed every figure independently rather than checking the author's working.
 - [] I tested each distractor for whether it is defensible in a sector or jurisdiction other than my own.
 - [] My findings go back to the author; I have not edited the item myself.
-

The reason to publish a lifecycle rather than assert quality is that a lifecycle can be audited and an assertion cannot. Anyone can say their items are written by experts. The question worth asking of any certification body is what happens to an item that the data says is broken — and whether the answer is a documented rejection with a retained record, or a quiet edit that nobody outside will ever see.

— Related

- [EXB-03 – Cognitive levels and item types](#) — the format specification these rules implement
 - [EXB-05 – Standard setting: modified Angoff and the pass mark](#) — where item difficulty enters the passing standard
 - [EXB-06 – Job-task analysis: where the blueprint gets its authority](#) — what tells the Assessment Lead where the bank should be deep
 - [EXB-07 – Examination security and integrity](#) — exposure, leakage and the protection of the bank
-

— Sources and standards

The statistics in §7 are standard classical test theory measures — the proportion correct, the upper/lower discrimination index and item-total correlation — described here in the Institute's own words. The worked figures in §7.4 are illustrative and were independently recomputed before submission. The certification framework is developed with reference to ISO/IEC 17024 personnel-certification principles; PCI is not accredited by any ISO/IEC 17024 accreditation body.

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EXB-05 · EXAM BLUEPRINT

Standard setting: modified Angoff and the pass mark

How a passing standard is decided, why 65 per cent is a configured mark and not yet a validated standard, and why candidates never compete with each other

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

A pass mark is not a round number chosen for pass-rate optics; it is a documented judgement about what a minimally competent professional must be able to do. This document explains how the PCL-AI passing standard will be set by a modified-Angoff standard-setting study — who sits on the panel, how the minimally competent candidate is defined, what happens in each round, and how the panel's recommendation becomes a published standard. It works the arithmetic through with illustrative figures, states plainly that 65 per cent is the platform's configured mark today while the definitive standard awaits the study, and explains why criterion-referenced scoring means no candidate's result depends on anyone else's. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

— About this document

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Standard setting: modified Angoff and the pass mark

In one paragraph. A pass mark is not a round number chosen for pass-rate optics; it is a documented judgement about what a minimally competent professional must be able to do. This document explains how the PCL-AI passing standard will be set by a modified-Angoff standard-setting study — who sits on the panel, how the minimally competent candidate is defined, what happens in each round, and how the panel's recommendation becomes a published standard. It works the arithmetic through with illustrative figures, states plainly that 65 per cent is the platform's configured mark today while the definitive standard awaits the study, and explains why criterion-referenced scoring means no candidate's result depends on anyone else's. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

Who this is for. Candidates who want to know what the pass mark actually represents; employers deciding what a PCL-AI pass evidences; and assessment professionals auditing whether the Institute's standard is set by a method or by a preference.

— 1. Where the pass mark comes from — the honest position

Two statements are both true, and the Institute publishes both because a candidate is entitled to know which one they are being measured against.

65 per cent is the platform's configured pass mark. It is what the examination applies today, it is what appears in the candidate handbook, and it is the figure against which a result is currently decided.

The definitive standard will be set by a modified-Angoff standard-setting study. Until that study reports, 65 per cent is a configured operating value — a defensible starting point, chosen in the same region as long-standing practice in professional certification — and not a validated standard derived from this examination's own content.

Those are different things. A configured value is an administrative setting. A validated standard is a documented judgement about competence, made by qualified people, on this examination's actual items, with a record that can be audited afterwards. Presenting the first as though it were the second is the specific misrepresentation this document exists to avoid.

— 2. What a passing standard is a judgement about

A pass mark answers a question that sounds simple and is not: **how much of this examination must a candidate answer correctly to demonstrate the competence the credential attests?**

Notice what the question is not. It is not how many candidates should pass — that is a norm-referenced question, and the Institute does not ask it. It is not what mark feels rigorous — that is aes-

thetics. And it is not what did we use last time — that is inheritance, which is how a great many pass marks in professional certification were actually arrived at.

The question is about the content. Standard setting answers it by having qualified people examine the actual items and judge, item by item, what a just-competent professional would do with them.

— 3. Criterion-referenced: why candidates do not compete

The PCL-AI is **criterion-referenced, not graded on a curve**. Everyone who demonstrates the required standard passes, however many others do. There is no quota, no fixed proportion who succeed or fail, and no ranking against other candidates.

The difference is not academic. Illustrative figures.

Consider two candidates who give **identical** answers to an identical form, sitting in different months.

Under a **norm-referenced** design in which, say, the top 30 per cent of each cohort passes, the outcome depends on the cohort. In a month when the candidate sits alongside an unusually strong group, identical performance ranks below the thirtieth percentile and fails. In a weaker month, it ranks above and passes. The candidate's competence has not changed; only the company they kept has.

Under the **criterion-referenced** design the Institute uses, identical performance produces an identical outcome, in every month, in every cohort, on every form assembled to the blueprint. The comparison is against a published standard, not against other people.

Three consequences follow, and they are worth stating because candidates ask about all three.

A resit clears exactly the bar a first sitting cleared. The standard does not rise because you failed once, and it does not fall because the cohort is small.

There is no advantage in sitting at a quiet time of year, and no disadvantage in sitting alongside a strong cohort.

The Institute publishes no target pass rate. A pass rate is an outcome of a criterion-referenced system, not an input to it. A certification body that manages its pass rate towards a number has stopped running a criterion-referenced examination, whatever its documentation says.

— 4. The minimally competent candidate

Everything in standard setting rests on one construct, and getting it wrong invalidates the whole study.

The **minimally competent candidate** is the person who is just good enough. Not the excellent candidate, not the average candidate, not the candidate the panellists would want to hire — the one who sits exactly at the boundary the credential is drawing. They would be safe to certify; anyone slightly weaker would not.

Panels find this genuinely hard, and predictably so. Experienced practitioners tend to picture a good professional, because that is who they work with and who they would recommend. Picturing a good professional and rating items against them sets the standard too high — and it does so invisibly, because every rating looks reasonable on its own.

The study therefore spends its first substantial block of time not rating anything, but building a shared, written description of this person: what a just-competent PCL-AI holder can do, what they would still need to check, what they would escalate, and what they would get wrong. That description is written down, agreed, and referred back to whenever ratings diverge. A panel that has not written it down is rating against several different people at once and does not know it.

— 5. Panel composition

A standard-setting panel is a measurement instrument, and its composition determines what it can measure.

The Institute's requirements for the PCL-AI panel:

Subject-matter expertise across the whole blueprint. Panellists between them must cover project accounting and finance, project management principles, and AI in project controls. A panel with no credible AI practitioner cannot rate Domain 13 items, and will systematically misjudge them in one direction or the other.

Familiarity with the target population. Panellists must know the people who will actually sit the examination — what a practitioner with three years' experience in any field typically knows and does not know. Expertise without that familiarity produces a standard set for a population that does not exist.

Diversity of region and sector. Project controls is practised differently in construction, energy, infrastructure, manufacturing, defence and technology delivery, and differently again across regions. A single-sector panel encodes its own sector's norms as the standard.

Independence. Panellists do not rate items they wrote, and the panel is chaired by someone who does not rate at all — the chair manages process, not content. Anyone with a commercial interest in the pass rate, including training providers, is excluded from rating.

Trained on the method. Panellists are trained on the task, practise on items outside the study, and demonstrate they understand the rating question before the study begins. Untrained panels produce ratings that look like data and are not.

Panel size and precise composition: [\[CONFIRM: standard-setting panel size and composition\]](#). The study date: [\[CONFIRM: date of the standard-setting study\]](#).

— 6. The method

The Angoff method takes its name from the psychometrician William Angoff. **Modified** distinguishes the multi-round, feedback-informed variants now in general use from the original single-round formulation.

6.1 The rating question

For each item, every panellist independently answers one question:

Out of 100 minimally competent candidates, how many would answer this item correctly?

The answer is a number from 0 to 100. It is **not** a judgement of whether the item is good, whether the panellist likes it, or whether the panellist would answer it correctly. It is an estimate about a specific imagined population — the one described in §4.

Two disciplines apply. Panellists rate against the written description of the minimally competent candidate, not against themselves. And because a four-option item has a chance floor of 25 per cent ($1 \div 4$), a rating materially below 25 flags the item rather than the candidate: it says the panellist believes minimally competent candidates would do worse than random guessing, which is a statement about a defective item.

6.2 Round one — independent

Panellists rate every item alone, without discussion. Independence matters: a first round contaminated by the loudest voice in the room produces one person's standard wearing a panel's clothes.

6.3 Feedback and discussion

The panel is then shown, item by item, the spread of its own ratings — the mean and the range, without attribution. Discussion focuses on the items where the spread is widest, because a wide spread means panellists were rating different people or reading the item differently, and both are fixable.

Where empirical data exists — the item statistics from [EXB-04 – How items are written, reviewed and retired](#) §7 — it is introduced at this point. Panellists see how candidates actually performed on the item. This is one of the two things that make the method "modified", and it is a substantial improvement: it anchors judgement without replacing it, and it catches the item that everyone on the panel thought was easy and no candidate could answer.

Discussion is not a negotiation towards consensus. Nobody is asked to change a rating, and a panellist who holds their original view after hearing the discussion is doing the task correctly.

6.4 Round two — revised

Panellists rate again, independently, having heard the discussion and seen the data. The round-two ratings are the ones that count.

6.5 Computing the cut score

For each item, the item cut score is the mean of the panellists' round-two ratings. The recommended cut score for the form is the mean of the item cut scores — equivalently, the sum of the item means divided by the number of items.

Because every scored item carries equal weight (see [EXB-03 – Cognitive levels and item types](#) §4.2), that figure is directly a percentage of the form.

— 7. Worked example

Illustrative figures. Five panellists, six items, round-two ratings. Ratings are the estimated percentage of minimally competent candidates answering correctly. Means are rounded to one decimal place; the final cut score to one decimal place.

Item	P1	P2	P3	P4	P5	Sum	Item cut score (mean)	Range
1	80	75	85	70	80	390	78.0	15
2	60	55	65	50	60	290	58.0	15
3	45	40	55	35	50	225	45.0	20
4	70	65	75	70	65	345	69.0	10
5	55	60	50	60	55	280	56.0	10
6	85	90	80	85	85	425	85.0	10
Total of item means							391.0	

Recommended cut score = sum of item cut scores ÷ number of items = 391.0 ÷ 6 = 65.2 %

Read the table as a panel chair would.

Item 6 is easy for this population — the panel expects 85 per cent of minimally competent candidates to get it. That is not a criticism. An examination needs items across the difficulty range, and an item this easy still contributes if it discriminates.

Item 3 is the one to discuss. Its ratings span 35 to 55, the widest range on the form, and it sits closest to the chance floor. A range that wide after a feedback round says panellists are still reading the item differently — which usually means the item is ambiguous, not that the panellists are careless. This is the item the chair sends back to the Assessment Lead for technical review, and if it is withdrawn the cut score is recomputed on the remaining items.

The result is a recommendation, not a decision. 65.2 per cent is what this panel, on this form, recommends. What happens next is §8.

— 8. From recommendation to published standard

A standard-setting study produces a number with error attached, and pretending otherwise is the most common way a good study produces a bad decision.

Measurement error is computed and stated. Every score carries a standard error of measurement — the spread of results the same candidate would produce on repeated equivalent sittings. It is computed from the form's own statistics, so it cannot be stated in advance of the form, and any body quoting one before its examination exists is quoting a guess. What the Institute commits to is that the error will be computed, published with the study, and taken into account by a **decision rule fixed before the study runs** — not selected afterwards once the panel's number is known, which is the point at which a technical adjustment becomes a thumb on the scale.

The decision belongs to governance, not to the panel. The panel recommends; the Chief Examiner and the Certification & Impartiality Committee decide. The decision may adopt the recommendation, or depart from it for reasons that are recorded in writing. What it may not do is depart from it silently, or for reasons related to the pass rate the standard would produce.

Everything is documented. The study report records the panel composition and how it was selected, the training given, the written description of the minimally competent candidate, both rounds of ratings, the discussion record, the empirical data introduced, the computed cut score, the meas-

urement error, the decision rule and the final decision with its reasons. That report is what makes the standard defensible in an appeal, and its existence is the difference between a standard and an assertion.

Forms are held to the same standard, not to the same raw score. A standard is a level of competence. If two forms differ in difficulty, holding the raw percentage constant would silently move the standard, so the raw passing score must be adjusted to keep the standard where it is. Today the platform applies a single configured percentage to every form. Once item statistics support it, any form-level adjustment the Institute adopts will be published in this document rather than applied quietly — because a candidate who discovers after the fact that forms were not equated has been told something untrue about comparability.

— 9. If the study moves the mark

The study may put the standard somewhere other than 65 per cent. That is the point of running it, and the Institute's commitments if it happens are stated now, before there is any incentive to word them conveniently.

The change is published, with the study's method, panel composition and reasoning.

It applies from a stated effective date, with notice. Candidates who have booked against a published standard sit against that standard.

It is never applied retrospectively to results already issued. A credential earned against the standard in force at the sitting remains earned.

It is published in either direction. A standard that moves down is published exactly as a standard that moves up. A body that only announces its increases is managing perception, not standards.

— 10. How standard setting goes wrong

Rating against the good candidate. The dominant failure mode, described in §4. It inflates the standard and it is invisible without a written description of the minimally competent candidate to rate against.

Choosing the number first. A panel convened to justify 65 per cent will find its way to approximately 65 per cent. The order matters: define the candidate, rate the items, compute the result, then decide — and publish the result even when it is inconvenient.

A panel that is really one person. Without independent first-round ratings and unattributed feedback, the most senior or most confident voice sets the standard and five signatures make it look like a panel.

Introducing empirical data too early. Show panellists the item statistics before they have formed their own judgement and they anchor on the data, which turns a criterion-referenced study into a norm-referenced one wearing a disguise.

Treating the cut score as exact. 65.2 per cent is not meaningfully different from 65.4 per cent, and a process that behaves as though it were is over-reading its own precision. This is what the measurement error in §8 exists to discipline.

Never revisiting it. A standard set once and inherited for a decade is a standard set for a discipline that no longer exists — a particular risk where Domain 13 content ages fastest.

Letting the pass rate lead. The moment a pass rate becomes a target, the standard has been abandoned. This is the failure that is hardest to detect from outside and easiest to justify from inside.

— 11. Checklist

For anyone auditing a standard-setting study — ours or another body's:

- [] Is there a written description of the minimally competent candidate, agreed before rating began?
 - [] Were panellists trained on the method and given practice items outside the study?
 - [] Was round one rated independently, before any discussion?
 - [] Was feedback unattributed, and focused on items with the widest spread?
 - [] Was empirical item data introduced after round one, not before?
 - [] Were panellists free to keep their original ratings after discussion?
 - [] Was the decision rule for measurement error fixed before the panel's number was known?
 - [] Does the report name the panel's composition, and does the composition cover the whole blueprint?
 - [] Were people with a commercial interest in the pass rate excluded from rating?
 - [] Is the study documented well enough that an appeal could be answered from it years later?
 - [] Is the resulting standard published, including if it moved downwards?
-

The reason to describe a standard-setting study in this much detail before running one is that the details are where standards are quietly lost. Any body can name the method; the method is not the safeguard. The safeguards are the written description of the borderline candidate, the independent first round, the decision rule fixed in advance, and the willingness to publish a number that is lower than the one you started with. When the PCL-AI study reports, this document is the specification it will be judged against.

— Related

- [EXB-01 – Examination blueprint – PCL-AI](#) — the configured pass mark and everything else the specification fixes
 - [EXB-04 – How items are written, reviewed and retired](#) — the item statistics introduced in the feedback round
 - [EXB-06 – Job-task analysis: where the blueprint gets its authority](#) — the study that defines the competence this standard is set against
 - [CER-07 – Results, scoring, appeals and complaints](#) — what a candidate does if they disagree with a result
-

— Sources and standards

The modified-Angoff method is named after the psychometrician William Angoff and is described here in the Institute's own words; no published text is reproduced. The illustrative ratings in §7 are invented for teaching and were independently recomputed before submission — they are not data from any study, panel or candidate population. The certification framework is developed with reference to ISO/IEC 17024 personnel-certification principles; PCI is not accredited by any ISO/IEC 17024 accreditation body.

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Founding-stage document · Version 1.0 · draft · Reviewed under PCI governance. PCI makes no claims of accreditation or recognition beyond what is true today.

EXB-06 · EXAM BLUEPRINT

Job-task analysis: where the blueprint gets its authority

The study that will fix the item counts, test the 40/40/20 weighting against evidence, and replace expert opinion with practitioner data

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

A blueprint cannot validate itself. A job-task analysis is the study that asks practitioners what the work actually consists of, how often each task is performed and how badly it goes when performed wrongly, and then links those tasks back to the examination's content. This document explains what a job-task analysis is, what the PCL-AI study will settle — including the item counts, which do not exist today — how the method works step by step, and how the resulting data converts into sampling weights. It works the linkage arithmetic through with illustrative figures and states what stands until the study reports. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

— About this document

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Job-task analysis: where the blueprint gets its authority

In one paragraph. A blueprint cannot validate itself. A job-task analysis is the study that asks practitioners what the work actually consists of, how often each task is performed and how badly it goes when performed wrongly, and then links those tasks back to the examination's content. This document explains what a job-task analysis is, what the PCL-AI study will settle — including the item counts, which do not exist today — how the method works step by step, and how the resulting data converts into sampling weights. It works the linkage arithmetic through with illustrative figures and states what stands until the study reports. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

Who this is for. Practitioners who may be asked to take part; employers deciding whether the credential measures the work their people actually do; and assessment professionals auditing the basis of the published blueprint.

— 1. The question a blueprint cannot answer about itself

EXB-01 – Examination blueprint – PCL-AI states what is examined and in what proportion. It is a careful document, built by experienced practitioners from a body of knowledge that took a great deal of work to assemble. And it cannot, on its own, answer the question that matters most:

How do you know the examination measures the job?

Expert judgement is the right starting point and it is not sufficient, for a reason that has nothing to do with the quality of the experts. Experts are unrepresentative by definition. They are more senior than the candidate population, they have worked in a particular set of sectors, they remember the discipline as it was when they learned it, and they systematically over-value the parts they find interesting. Ask ten excellent controls professionals what a credential should test and you will get ten defensible answers, no two of which allocate weight the same way.

A **job-task analysis** is the instrument that replaces that argument with data. It is the difference between a syllabus that experts believe in and a syllabus that can be shown to correspond to the work.

— 2. What a job-task analysis is

A job-task analysis is a structured study of a role, with three components.

A task inventory — a comprehensive list of the discrete tasks that make up the role, written at a consistent level of granularity and in the language practitioners actually use.

A validation survey — the inventory put to a large, representative sample of practitioners, who rate each task on defined scales. The two scales that matter are **frequency** (how often do you perform this?) and **criticality** (how serious are the consequences when it is performed badly?).

A linkage — an explicit mapping from tasks to the knowledge and skill required to perform them, and from there to the examination's content structure.

The output is not an opinion about what should be examined. It is a defensible statement, traceable to practitioner responses, about what the role consists of and which parts of it carry the most weight.

— 3. What the PCL-AI study will settle

Five things, of which the first is the one this series repeats most often.

3.1 The item counts

No item count exists for the PCL-AI. Not published, not configured, not privately assumed. Anyone quoting one is quoting something the Institute has not produced.

Item counts will be confirmed by the job-task analysis, in combination with a reliability target. The logic runs in that order and not the other way round: the study establishes what proportion of the role each content area represents; the reliability requirement establishes how many items are needed to measure each area with acceptable precision; the total follows from both. A sub-score computed from a handful of items carries a wide margin of error, which is why an area's item count is not simply its share of the blueprint multiplied by a number chosen in advance.

Setting the total first and reverse-engineering a justification is the failure mode. It is also, in the Institute's judgement, how a large number of professional examinations arrived at their item counts.

3.2 Whether the 40/40/20 weighting survives contact with evidence

The current weighting is a considered judgement, defended in [EXB-02 – Domain weightings and the 40/40/20 rule](#) §4 on grounds the Institute is willing to argue for. It is not yet an evidenced one.

The study will produce weighted task data that maps onto the three groups. If that data supports the current weighting, the weighting stands and is thereafter evidenced rather than merely argued. If it does not, the weighting changes — published as a major version of [EXB-01](#), with an effective date and a notice period. The Institute has committed to that in advance, in public, which is the only point at which such a commitment costs anything.

3.3 The sampling density within groups

[EXB-01](#) §4 states that weights attach to groups, not to individual domains. The study will show which domains within each group carry the most weighted task activity, and therefore where sampling should be denser. That governs form assembly and it governs where the item bank needs to be deep — the commissioning input described in [EXB-04 – How items are written, reviewed and retired](#) §3.

3.4 The description of the minimally competent candidate

The standard-setting study in [EXB-05 – Standard setting: modified Angoff and the pass mark](#) depends entirely on a written description of the just-competent professional. That description should be derived from the task data, not composed by the standard-setting panel from memory. A panel writing its own description of the borderline candidate is doing two studies at once and doing neither properly.

3.5 Whether a per-group hurdle is warranted

[EXB-02 §6](#) sets out the honest limitation of a compensatory scoring model, and says the question of a per-group minimum will be put to the job-task analysis. If the data shows that competence in one group cannot in practice substitute for competence in another — that they are distinct capabilities rather than facets of one — that is an argument for a hurdle. If it shows they covary strongly, it is an argument against. This is exactly the sort of question that opinion cannot settle and data can.

— 4. The method

4.1 Define the role and its boundary

The study begins by defining who is in scope, because a task inventory for "project controls" that includes everyone from a document controller to a portfolio director describes no role at all.

The PCL-AI attests competence at Leader level in the integrated discipline of project controls, cost engineering and project finance, and it is open to anyone with three years of professional experience in any field. The boundary must accommodate both facts: the study describes the role the credential attests, across sectors and regions, without narrowing to any single organisation's job description.

4.2 Build the task inventory

Three inputs, in this order:

Document review — the existing body of knowledge, published role descriptions, competency frameworks, and the artefacts controls professionals actually produce. This generates a first draft, and it is only a draft, because documents describe what organisations say the job is.

Practitioner workshops — facilitated sessions with working professionals across sectors and regions, who add what the documents missed, delete what no longer happens, and correct the language. This is where most of the value is created, and it is the stage most often skipped.

Editing to a consistent grain — the hardest editorial task in the study. "Manage cost" is too coarse to rate. "Open the cost report spreadsheet" is too fine. Tasks must sit at a level where a practitioner can answer both rating questions meaningfully, and every task in the inventory must sit at roughly the same level, or the ratings are not comparable.

The inventory must be **comprehensive and non-overlapping**. A task that appears twice in different words gets double-counted in the weighting, which quietly inflates whichever area it sits in.

4.3 The validation survey

The inventory is put to a large sample of practitioners. Each task is rated on defined scales.

Frequency — how often the respondent performs the task, on a scale from never to daily.

Criticality — how serious the consequences are when the task is performed incorrectly, from negligible to severe.

A third question is worth asking and is often omitted: **where the capability was acquired** — on the job, through formal study, or not yet at all. That answer tells the Institute where an examination adds most value, because assessment adds least where experience reliably teaches the skill and most where it does not.

Respondents also supply role, sector, region and years of experience, so that responses can be analysed by segment. A weighting that holds across sectors is a stronger result than an average that conceals two opposing patterns.

Scale design and survey construction: [\[CONFIRM: job-task analysis panel composition and survey design\]](#).

4.4 Sampling

The survey's authority rests entirely on who answered it, and this is where job-task analyses most often fail quietly.

A sample dominated by one sector, one region or one seniority band produces a weighting for that group and presents it as a weighting for the discipline. The Institute's requirements are that the sample spans sectors, spans regions, spans experience levels within the population the credential serves, and is large enough that segment-level analysis is possible rather than merely reported.

Response rates and any known non-response bias will be reported with the results. A study that publishes its weighting and not its sample is asking to be taken on trust in the one respect where trust is least warranted.

4.5 Analysis: from ratings to weights

Frequency alone is the wrong model, and it is the model most often used.

A task performed daily with trivial consequences matters less to a credential than a task performed twice a year that destroys a project when it goes wrong. Certification exists to protect against the second. So the weighting combines both scales — typically as a product, so that a task must score on both dimensions to carry weight, and a task rated zero on criticality drops out however often it is performed.

4.6 Worked example of the linkage arithmetic

Illustrative figures. Five tasks, mean ratings on 0–4 scales, standing in for a full inventory of several hundred. Products are shown to two decimal places; shares are of the total weighted inventory and are rounded to two decimal places.

Task (illustrative)	Frequency	Criticality	Weight = F × C	Share
Reconcile the project cost ledger to the general ledger at period end	3.4	3.8	12.92	27.73 %
Compute and interpret cost and schedule performance indices	3.6	3.5	12.60	27.04 %
Prepare an estimate at completion and justify the method chosen	2.8	3.9	10.92	23.44 %
Validate a machine-generated cost forecast before it is issued	1.9	3.7	7.03	15.09 %
Price a bill of quantities remeasurement	1.2	2.6	3.12	6.70 %
Total			46.59	100.00 %

Substitutions: $3.4 \times 3.8 = 12.92$; $12.92 \div 46.59 = 0.2773 = 27.73 \%$. The remaining rows follow the same two steps.

Now read the fourth row, because it is the reason criticality is in the model at all.

Validating a machine-generated forecast has the lowest frequency of the four substantive tasks (1.9) and the second-highest criticality (3.7). On frequency alone it would rank near the bottom of the inventory and attract minimal sampling. Weighted by criticality it takes 15.09 per cent — because it is not yet a daily task for most practitioners, and when it is done badly a fluent, confident, wrong forecast goes to a board.

That single row is the empirical form of the Institute's position on AI: the weight belongs to the consequence, not to the frequency. It is also a concrete illustration of how a defensible 20 per cent group weight could emerge from data rather than from conviction — which is precisely what the study exists to establish or refute.

The final step is **linkage**: each task is mapped to the knowledge areas required to perform it, task weights are aggregated up through knowledge areas to domains to groups, and the resulting group shares are compared against the published 40/40/20. Where a task requires knowledge from more than one area, its weight is apportioned, and the apportionment rule is fixed before the analysis is run.

4.7 Review and adoption

The analysis goes to a review panel drawn from outside the study team, then to the Standards Committee and the Certification & Impartiality Committee for adoption. The full study — inventory, scales, sample composition, response rates, analysis and linkage — is published alongside whatever blueprint change it produces, including the case where it produces none.

Study completion: [\[CONFIRM: job-task analysis completion date\]](#).

— 5. What stands until the study reports

The Institute is founding-stage, and the honest position is this.

The published blueprint is **expert-derived and internally coherent**: thirteen domains, sixty-one knowledge areas, three groups, 40/40/20, built from a body of knowledge developed to handbook depth. It is a defensible specification, it is published in full, and it is the specification every form is assembled to today.

It is **not yet practitioner-validated**. That is a real limitation and it is stated here rather than left for a reviewer to discover. The certification framework is developed with reference to ISO/IEC 17024 personnel-certification principles, of which a validated relationship between the examination and the role is one; PCI is **not accredited** by any ISO/IEC 17024 accreditation body, and completing this study is one of the things that would have to be true before accreditation could sensibly be sought.

Until it reports: the weighting stands as published, the item counts remain unstated, and nobody at the Institute quotes a number for them.

— 6. How a job-task analysis goes wrong

A sample that represents one sector. The most common and most consequential defect. The weighting that results is real, precisely measured, and about the wrong population.

Ratings from experts instead of practitioners. A job-task analysis surveyed to a panel of senior experts reproduces the expert opinion it was supposed to replace, now wearing the authority of a study.

Frequency-only weighting. Under-serves exactly the low-frequency, high-consequence tasks a credential exists to protect against — the fourth row of §4.6.

An inventory at inconsistent grain. Coarse tasks absorb weight that fine tasks fragment. The area of the inventory written most carefully ends up looking least important.

Overlapping tasks. Double-counts weight into whichever area the duplication sits in.

Running the study and then ignoring it. The failure that destroys the credibility of every other stage: a study that produces an inconvenient weighting, followed by a blueprint that does not change and a report that is not published. The public commitment in §3.2 exists to make that outcome costly.

Treating it as a one-off. A task inventory describes the work at a moment. The Institute's discipline is changing fastest in exactly the area weighted at 20 per cent, so a study left unrepeated for a decade describes a job nobody has any more.

— 7. Cadence

A job-task analysis is repeated on a defined cycle rather than when someone notices it is stale. Two pressures set the interval.

The **structural content** of project controls — accounting, earned value, contracts, scheduling, risk — changes slowly, over years, and does not need frequent re-study.

Domain 13 content changes inside a single certification cycle. A candidate certified for three years may hold a credential whose AI content was validated against practice that has since moved substantially. The Institute's approach is therefore a full study on the long cycle, with a lighter tar-

geted refresh of the AI-related task inventory between full studies — which is the same reasoning that puts a mandatory AI-currency component into recertification.

— 8. Checklist

For a practitioner invited to take part:

- ☐ Rate frequency against what you actually do, not what your job description says.
- ☐ Rate criticality against the consequence of doing it badly, not the consequence of not doing it.
- ☐ Rate the whole inventory, including tasks you never perform — a genuine zero is data.
- ☐ Answer the segment questions honestly; they are what allow the analysis to detect sector bias.

For anyone auditing a job-task analysis — ours or another body's:

- ☐ Is the task inventory published, not just its conclusions?
- ☐ Is the sample composition published, by sector, region and experience?
- ☐ Are response rates and known non-response bias reported?
- ☐ Does the weighting model include criticality, or only frequency?
- ☐ Is the task-to-content linkage published, with the apportionment rule for multi-area tasks?
- ☐ Were the weights compared against the existing blueprint, and was the comparison published even where it disagreed?
- ☐ Is there a stated cadence for repeating the study?

Publishing a blueprint before the study that validates it is a choice with a cost: it invites exactly the question this document opens with, and the Institute cannot yet answer it with data. The alternative — to publish nothing until everything is validated — would leave candidates preparing against a syllabus nobody had seen. The position taken here is to publish the specification, name precisely what it does not yet rest on, and commit in advance to changing it if the evidence says so. That commitment is only worth anything because it was made before the evidence arrived.

— Related

- [EXB-01 – Examination blueprint – PCL-AI](#) — the specification this study will validate or change
- [EXB-02 – Domain weightings and the 40/40/20 rule](#) — the weighting that will be tested, and the hurdle question the study will settle
- [EXB-05 – Standard setting: modified Angoff and the pass mark](#) — the study that depends on this one's description of the borderline candidate
- [CMP-03 – PCL-AI: the fourteen competencies](#) — the competency set the task inventory must be reconcilable with

— Sources and standards

The method described here is standard job-task analysis practice in professional certification, set out in the Institute's own words; no published text, instrument or scale is reproduced. The figures in §4.6 are illustrative, invented for teaching, and independently recomputed before submission — they are not data from any survey, practitioner or organisation. The certification framework is developed with reference to ISO/IEC 17024 personnel-certification principles; PCI is not accredited by any ISO/IEC 17024 accreditation body.

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— Status and version

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EXB-07 · EXAM BLUEPRINT

Examination security and integrity

How the item bank is protected, what proctoring actually does, why a result under review shows no score, and what candidates are obliged to do

VERSION 1.0 · DRAFT
2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.
AI proposes. The professional disposes.

— Summary

Examination security exists to protect one thing: that a pass means the same for everyone who earns it. This document sets out how the PCL-AI item bank is protected and how leakage is detected, what the proctoring arrangements actually do and honestly cannot do, how the secure exam client is designed so that the server rather than the candidate's machine owns the clock and the scoring, what evidence is captured and how long it is kept, and why a result placed under integrity review displays no score and no pass or fail outcome. It states candidate obligations plainly, and it says what is not misconduct. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

— About this document

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Examination security and integrity

In one paragraph. Examination security exists to protect one thing: that a pass means the same for everyone who earns it. This document sets out how the PCL-AI item bank is protected and how leakage is detected, what the proctoring arrangements actually do and honestly cannot do, how the secure exam client is designed so that the server rather than the candidate's machine owns the clock and the scoring, what evidence is captured and how long it is kept, and why a result placed under integrity review displays no score and no pass or fail outcome. It states candidate obligations plainly, and it says what is not misconduct. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

Who this is for. Candidates preparing to sit; employers relying on the credential; and anyone auditing whether the Institute's integrity claims describe a system or an intention.

— 1. What security is actually protecting

Not the Institute's revenue, and not its dignity. The thing being protected is **the value of every credential already issued**.

A candidate who obtains a pass by cheating has not only obtained something they are not entitled to; they have taken a small amount of meaning away from every other holder's credential, and from every employer who relied on it. That is why examination security is a duty owed to certificants rather than a defensive measure against candidates, and why the Institute publishes its arrangements instead of leaving them vague.

There is a second, less obvious protection. A rigorous integrity process protects the **innocent** candidate — the one whose broadband failed, whose flatmate walked through the room, or whose face was momentarily lost by a camera. A system that cannot distinguish an anomaly from misconduct will eventually punish someone who did nothing wrong, and the arrangements below are designed with that failure in mind as much as the other.

— 2. Protecting the item bank

The item bank is the examination's most valuable and most vulnerable asset. Five arrangements protect it.

2.1 Published and live content are separated absolutely

No live examination item is ever published. Not in a sample paper, not in a training partner's course, not in a marketing document, not on request.

No published item is ever scored. The sample items in [EXB-08 – Sample items with rationale](#) are authored for publication, marked as study material, and held in a separate store from the live bank. They are written to the same blueprint, the same format and the same cognitive levels — which is what makes them useful — and they are not, and never become, live content.

The separation runs in both directions on purpose. A body that publishes "retired" live items is training candidates on the exact material its bank was built from, and a body that promotes a published item into the bank has published its own answer key.

2.2 Access is controlled and logged

Access to live item content is restricted to the roles that require it — the Chief Examiner and Assessment Lead, the subject-matter experts reviewing a specific item, and the systems that assemble forms. Privileged actions in the platform are written to an audit log. Nobody has standing access to the whole bank because their job title is senior.

2.3 Exposure is tracked and forms are assembled, not fixed

Every use of an item on a form is recorded. There is no single fixed paper: each form is assembled to the blueprint from the bank, so two candidates sitting at the same time are not answering the same set of items in the same order. An item used beyond its tracked exposure threshold leaves the scored bank whether or not any leak is suspected — see [EXB-04 – How items are written, reviewed and retired](#) §10.

2.4 Leakage is detected statistically, not only by report

Waiting to be told about a leak is not a control. The bank's own statistics are the detector.

An item whose difficulty drifts steadily downwards — more candidates answering correctly, with no change in the population and no change to the item — is behaving exactly as a leaked item behaves. So is an item whose discrimination collapses while its difficulty falls, which is the signature of an item that weak candidates have memorised. Continuous monitoring of item statistics ([EXB-04](#) §9) is therefore a security control as much as a psychometric one.

Response patterns are also examined: implausibly fast correct responses, answer sequences shared across unconnected candidates, and clusters of candidates performing far better on a subset of items than on the rest of the form.

2.5 What happens on a confirmed or suspected leak

Retirement is **immediate and does not wait for proof**. An item under credible suspicion leaves the scored bank at once, because the cost of retiring a sound item is small and the cost of scoring a leaked one is paid by every candidate on the form.

Results already issued are not disturbed by a retirement. A candidate scored against the bank version in force at their sitting keeps that result. The one exception is a confirmed mis-key, where affected results are rescored and every affected candidate is notified — including those whose result improves and those for whom nothing changes.

2.6 Reconstructability

Every scored attempt records the bank version and the answer-key version it was scored against, and an immutable score snapshot is written at submission. A result can therefore be reconstructed exactly as it was computed, years later, even if items have since been edited or retired. This is what makes a late appeal answerable with evidence rather than with recollection — see [CER-07 – Results, scoring, appeals and complaints](#).

— 3. Delivery: what proctoring does

The PCL-AI is delivered online under live proctoring. Delivery arrangements, booking and the technical requirements are governed by the published policies and summarised in [CER-06 – Online proctoring and the secure exam client](#); what follows is the integrity design.

3.1 Before the session starts

Identity verification. Valid government-issued photo identification is required, and the name on the booking must match it exactly, character for character. Name mismatch is the most avoidable cause of exam-day failure and the easiest to fix a week in advance.

Room scan. The candidate shows the workspace and its surroundings. The purpose is to establish that the space is clear of notes, devices and second screens before the examination opens, not to inspect a home.

System check. The candidate's machine, camera, microphone, displays and network are checked against requirements. This is run on the actual machine, network and room to be used, well before the day.

Consent. What is captured, why, and how long it is kept is disclosed and consented to before capture begins.

3.2 The launch window and the launch code

The session opens **15 minutes before** the booked time, with a **30-minute grace period** after it. Outside that window the session does not open.

The portal hands the desktop client a **short-lived, single-use launch code** — not a reusable credential. The code is redeemed once, against the Institute's own service, and is worthless afterwards. This matters for a specific reason: a stolen bearer token is a standing key to a candidate's account, and a redeemed single-use code is nothing at all.

3.3 The server owns the clock and the scoring

Two design decisions that a candidate cannot see and that determine whether the examination can be manipulated at all.

The clock is the server's. The client does not decide how much time remains. A heartbeat to the server returns the canonical remaining seconds, and a forced submission at expiry is driven by the server, not by the candidate's machine. Tampering with a local clock, suspending a process or disconnecting the network does not create time.

Scoring is never performed on the candidate's machine. The client displays items and transmits responses. It does not hold the answer key, and it does not compute a result. There is nothing on the candidate's machine worth attacking for a better score.

The client is pinned to the Institute's own host. It will only talk to an allowlisted service over HTTPS. A launch instruction pointing it somewhere else is ignored and the client refuses to start, which removes an entire category of attack in which a candidate is directed to a service that will score them generously.

3.4 Kiosk lockdown, and what it honestly cannot do

The desktop client applies user-space lockdown: it holds the foreground, blocks common hotkeys and clipboard use, watches for prohibited processes, detects additional displays, and flags remote-desktop and virtual-machine indicators.

It runs as an ordinary application, not as part of the operating system, and it says so. It cannot block the operating system's own secure attention sequence. It cannot see a second computer sitting on the desk, a phone under the table or a person out of shot. It cannot certify that a room is empty.

That honesty is deliberate, and it has a practical consequence: **the client's job is to detect and record, not to make cheating impossible.** No consumer-hardware proctoring system can make cheating impossible, and a body claiming otherwise is either mistaken or selling something. What the arrangements do is raise the effort required, capture what happened, and put a trained human in front of the evidence.

3.5 What the session records

Integrity signals are time-stamped, streamed to the Institute's service and compiled into a session record. The categories are: session lifecycle events; identity verification outcome and room-scan completion; focus loss and regain; prohibited processes; display connection and removal; virtual-machine and remote-desktop indicators; blocked clipboard and hotkey attempts; camera signals such as no face detected, multiple faces or prolonged absence; audio signals such as speech or loud sound; network drops and restores; missed heartbeats and client crash-resume; answer saves and submission; and any manual flag raised by a proctor.

Each signal carries a severity from informational through low, medium and high to critical.

3.6 An automated flag is not a verdict

This is the most important sentence in the section, and it describes how the platform is actually configured.

Proctoring and identity signals are audit-only by default. They are recorded, they are compiled into the session record, and they inform human review. Automatic invalidation of a result on the strength of a proctoring signal or an identity check is an optional configuration that is switched off by default — which means, in the arrangement the Institute operates, no candidate's result is destroyed by an algorithm.

A cat walking across a desk, a delivery at the door, a network drop in a country with unreliable connectivity — all of these generate signals, and none of them is misconduct. The system's job is to notice; a person's job is to decide.

— 4. Evidence, privacy and retention

Proctoring evidence is personal data captured under consent for a single purpose: deciding whether an examination session was conducted properly.

Access is need-to-know. Session evidence is available to the people conducting a review, not to administrators generally.

It is used for that purpose only. It is not used for research, marketing or product development, and it does not inform any assessment judgement about the candidate's answers.

It is retained for a defined period and then the artefacts are purged, with the metadata record kept for audit. The retention period is [\[CONFIRM: retention period for proctoring evidence\]](#).

Accommodation information is handled separately. Where a candidate has an approved adjustment — extra time, agreed rest breaks, assistive technology — the proctoring arrangements are configured in advance so that a legitimate break is never flagged as an irregularity. The proctor knows an arrangement exists and what is needed to run the session, not why it was granted, and the information is kept apart from assessment judgements.

— 5. The held-result rule

A result under integrity review shows no score, no pass or fail outcome, and no credential. Not a provisional score. Not a score marked "subject to review". Nothing.

The candidate sees a clear message: the responses were submitted successfully, the result is on hold pending an examination integrity check, and the Institute will notify them through the portal when the review concludes.

The rule is enforced in two independent places — the server redacts the score before it leaves the service, and the desktop client's presentation layer refuses to display a score for a held submission regardless of what any payload contains. Two enforcement points for one rule is deliberate: a rule this consequential should not depend on a single piece of code being correct.

5.1 Why the rule is absolute

A number, once seen, cannot be unseen. Show a candidate 71 per cent and then invalidate the attempt, and you have created a grievance that no amount of subsequent process will settle. The candidate will reasonably feel something was taken from them.

A provisional score prejudices the review. Once a figure has been disclosed, everyone involved is deciding whether to withdraw a pass rather than whether the session was sound. Those are different questions and they do not produce the same answers.

Partial disclosure invites reconstruction. A candidate who is shown a domain breakdown but not a total, or a score but not an outcome, can often infer the rest — and a candidate who can infer the outcome has been told it.

It protects the candidate as much as the credential. A held result that resolves in the candidate's favour is released as a normal result, and nothing about the review is visible on the credential or the verification record. Had a provisional score been shown and then confirmed, the episode would follow the candidate as an explanation they had to give for years.

5.2 What a hold is not

A hold is **not a finding of misconduct**, and it is not an accusation. Holds arise from technical irregularities and process anomalies as well as from conduct questions. Most resolve without any adverse finding at all.

— 6. How an integrity review works

Fair process is the point, and its elements are not negotiable.

The candidate is told what is alleged, specifically enough to respond to. "An irregularity was detected" is not a case a person can answer.

The candidate is given a genuine opportunity to respond, in writing, with time to prepare, and may provide their own account and evidence.

The decision is taken by people with no conflict of interest — not by the proctor who raised the flag, and not by anyone with an interest in the outcome.

The evidence is examined, not summarised. A severity label is a pointer to a recording, not a substitute for looking at it.

Outcomes are proportionate, and range from no action, through a technical re-sit at no fault to the candidate, to invalidation of the result and, in the most serious cases, loss of the right to resit or withdrawal of an existing certification.

Adverse decisions carry a right of appeal, reviewed independently of the people who made the original decision. Raising an appeal never prejudices a candidate's standing. The procedure is in [CER-07 – Results, scoring, appeals and complaints](#).

— 7. Candidate obligations

Stated positively, because a list of prohibitions is easy to read and hard to follow.

Sit your own examination, and let nobody else sit any part of it. Impersonation is misconduct in both directions — the person who sits and the person who is sat for.

Present genuine identification in the name the booking is made in.

Work alone, with nothing but the interface. No notes, no reference material, no formula sheet, no second device, no second screen, no assistance from another person, and **no AI tool of any kind**. The examination assesses whether you can do this without help; using an assistant to answer it produces a credential that attests something untrue about you.

Follow the proctor's instructions throughout the session.

Keep the content confidential, during and after. Examination items are secured content. Reconstructing them from memory, sharing them in a study group or a forum, soliciting them from someone who has sat, or selling or buying them is misconduct however informal the setting and however long afterwards. This is the obligation candidates most often breach without intending to — a well-meaning "here's what came up" post in a study group is a breach, and it damages the credential the poster has just earned.

Report a disruption immediately, to the proctor during the session and to the Institute as soon as it ends, while the detail is fresh. A contemporaneous report from the candidate is the single most useful piece of evidence in a technical review, and it is nearly always in the candidate's favour.

— 8. What is not misconduct

Candidates worry about the wrong things, so the Institute says this explicitly.

Looking away from the screen to think is not misconduct. Speaking aloud while working through a calculation is not misconduct. A network drop is not misconduct. Somebody entering the room without your knowledge is not misconduct on your part. Requesting an accommodation has no bearing on how an examination is marked, in either direction, and it does not increase scrutiny. Using approved scratch working, where the delivery arrangements permit it, is not misconduct.

If something goes wrong, say so at the time. Candidates who report a problem immediately are in a substantially better position than candidates who say nothing and hope.

— 9. How this goes wrong

Believing that lockdown equals security. Kiosk software is one layer, it runs in user space, and a body that treats it as the whole control has stopped thinking. Detection, statistics, human review and the integrity of the item bank all matter more.

Automating the verdict. The efficient design is to let a severity threshold invalidate results automatically. It is also the design that will eventually destroy an innocent candidate's result and be unable to explain why. The Institute's platform defaults to audit-only for exactly this reason.

Showing a provisional score. Almost every body that does this does it to be helpful, and it is the single most damaging thing that can be done to a review's fairness.

Publishing live items as practice material. Usually requested with good intentions by a training partner who wants realistic preparation. It converts the examination into a memory test and cannot be undone.

Treating security as a candidate-relations problem. Weak enforcement to avoid complaints transfers the cost onto every honest candidate, silently, and the people who paid it never find out.

Retaining evidence indefinitely. Proctoring evidence is intrusive personal data captured for one purpose. Keeping it because storage is cheap is a privacy failure that the passage of time makes worse, not better.

Vague allegations. "An irregularity was detected in your session" gives a candidate nothing to answer and makes a fair process impossible. Specificity is a fairness requirement, not a courtesy.

— 10. Checklist

For a candidate, before the day:

- [] The name on my booking matches my photo identification exactly, character for character.
- [] I have run the system check on the actual machine, network and room I will use — not a different one.
- [] Any accommodation I need is requested and approved in advance, not raised on the day.
- [] My workspace will be clear of notes, devices and second screens, and second monitors are disconnected.

- [] I know the launch window opens 15 minutes before my booked time and closes after the 30-minute grace period.
- [] I have read the conduct rules and understand that no AI tool of any kind may be used.
- [] I know that if anything goes wrong I tell the proctor immediately and report it to PCI the same day.
- [] I understand that examination content is confidential during and after the sitting.

For anyone auditing an examination body's integrity arrangements:

- [] Are published and live items provably separate, in both directions?
- [] Is exposure tracked, and does over-exposure retire an item without a leak being proven?
- [] Is leakage detected statistically, or only when someone reports it?
- [] Can any historical result be reconstructed exactly as it was scored?
- [] Is scoring performed server-side, with the clock owned by the server?
- [] Do automated proctoring signals inform human review, or do they decide?
- [] Is a result under review displayed without any score or outcome?
- [] Is the allegation put to the candidate specifically enough to answer?
- [] Is the deciding body independent of the person who raised the flag?
- [] Is proctoring evidence retained for a defined period and then purged?

Every control described here is a cost — to the Institute, and sometimes to a candidate's convenience. The reason to carry it is that a credential is a claim made to third parties who cannot verify it themselves. An employer relying on a PCL-AI is relying on the proposition that the person in front of them sat the examination, alone, without assistance, against items nobody had seen. Everything above exists to make that proposition true, and publishing it is how the Institute invites being held to it.

— Related

- [EXB-04 – How items are written, reviewed and retired](#) — exposure tracking, item statistics and retirement, from the psychometric side
- [EXB-08 – Sample items with rationale](#) — the published study items, and the separation that keeps them apart from the bank
- [CER-06 – Online proctoring and the secure exam client](#) — the candidate-facing delivery arrangements and technical requirements
- [CER-07 – Results, scoring, appeals and complaints](#) — how a held result is resolved and how an adverse decision is appealed
- [ETH-01 – The PCI code of ethics and professional conduct](#) — the professional obligations that outlast the examination

— Sources and standards

The delivery, proctoring and held-result arrangements described here are the Institute's platform configuration and secure-exam client design as verified in August 2026. The certification frame-

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EXB-08 · EXAM BLUEPRINT

Sample items with rationale

Sixteen original study items across all thirteen domains, with every option explained and every calculation shown

VERSION 1.0 · DRAFT

2026-08-04

PROJECT CONTROLS INSTITUTE GLOBAL, INC.

AI proposes. The professional disposes.

— Summary

Sixteen original sample items covering all thirteen PCL-AI domains, written to the published blueprint in the examination's format and cognitive levels. Every item carries a full rationale that explains why the key is the key and what specific practitioner error each distractor represents, and every calculation shows its substitution. A closing section reads several of the items as an item writer would, so that candidates can see what a well-constructed item is doing to them. These are study items, written for publication and held separately from the live examination bank; no live examination content is ever published. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

— About this document

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Sample items with rationale

In one paragraph. Sixteen original sample items covering all thirteen PCL-AI domains, written to the published blueprint in the examination's format and cognitive levels. Every item carries a full rationale that explains why the key is the key and what specific practitioner error each distractor represents, and every calculation shows its substitution. A closing section reads several of the items as an item writer would, so that candidates can see what a well-constructed item is doing to them. These are study items, written for publication and held separately from the live examination bank; no live examination content is ever published. It applies to the PCL-AI only; no examination blueprint exists yet for PFL-AI or PML-AI.

Who this is for. Candidates preparing for the PCL-AI; training providers building practice material; and anyone who wants to see what the Institute means when it says its items assess judgement rather than recall.

— 1. What these items are, and what they are not

They are study material. Every item below was written for publication. They are authored to the same blueprint, the same four-option single-best-answer format and the same cognitive levels as the examination, which is what makes them useful preparation.

They are held apart from the live bank, permanently. No item published here is or will become a scored item, and no scored item is or will be published here. That separation is absolute and it runs in both directions — see [EXB-07 – Examination security and integrity](#) §2.1. A certification body that publishes its live items has converted its examination into a memory test, and one that promotes its published items into the bank has published its own answer key.

No live examination content is ever published, in a sample paper, a course, a marketing document or on request.

The number of items here is not an item count. Sixteen is how many fit usefully in a published document. **The PCL-AI has no published item count;** item counts will be confirmed by a formal job-task analysis — see [EXB-06 – Job-task analysis: where the blueprint gets its authority](#).

The mix here is not the form's specification either. These sixteen are distributed for domain coverage — one item in each of the twelve domains outside the AI group, and four in Domain 13 — so that a candidate can see the whole frame. A live form is assembled to the published 40/40/20 group weighting described in [EXB-02 – Domain weightings and the 40/40/20 rule](#), which is a different distribution.

— 2. How to use them

Work each item completely before reading its rationale. Write down your answer and, more importantly, write down the intermediate figure or the position you established on the way to it. Then read the rationale in full, including the options you did not choose.

The rationale is the part that teaches. Getting an item right for the wrong reason is worth almost nothing in preparation, and the only way to detect it is to compare your reasoning against the explanation rather than your letter against the key.

When you get one wrong, identify **which** distractor you chose and **which named error** it corresponds to. Every distractor below was built from a specific mistake practitioners actually make. Knowing which one you made is the diagnostic; knowing that you scored twelve out of sixteen is not.

— 3. Reading the tags

Each item is tagged [Domain · Knowledge area · Cognitive level] — for example [D6 · KA 6.3 · Analysis]. The cognitive levels are defined in EXB-03 – Cognitive levels and item types §2.

Across these sixteen items: two are Recall, five are Application and nine are Analysis. That is broadly the shape the examination intends — application-weighted, with Recall a genuine minority — though the exact proportions on a live form form part of the examination specification and are not published as a fixed mix.

All monetary figures are illustrative and expressed in USD. All periods, rates and quantities are illustrative. No item below describes a real project, organisation or person.

— 4. Group 1 — Project accounting & finance (Domains 1-4)

Item 1 [D1 · KA 1.3 · Application]

Illustrative figures.

At the 31 March cut-off, a subcontractor has completed work on site independently valued at USD 180,000. The subcontractor has submitted invoices totalling USD 120,000. The cost ledger records invoiced amounts only. What treatment does the March cost position require?

- **A.** Accrue USD 60,000 as cost incurred and not yet invoiced.
- **B.** Make no entry: the cost is recognised when the subcontractor's invoice is received.
- **C.** Accrue USD 180,000 in addition to the USD 120,000 already recorded.
- **D.** Recognise a provision of USD 60,000 for a liability of uncertain amount.

Key: A.

Work performed in a period is a cost of that period, whatever the invoicing has done. The accrual is the gap between value of work executed and value invoiced:

$$180,000 - 120,000 = 60,000$$

B is the cash-basis instinct, and it is the single most common cause of a project that appears to be under-running at every cut-off and over-running at the final account. It defers cost into the wrong period and flatters the current one.

C double-counts. Accruing the full 180,000 on top of the 120,000 already in the ledger records 300,000 of cost for 180,000 of work.

D confuses two adjacent concepts. A provision is recognised for a liability of uncertain timing or amount. Here the work is done, the value is measured and the obligation is certain — that is an accrual. Getting this boundary wrong is not merely presentational: provisions attract different disclosure and different scrutiny.

Item 2 [D2 · KA 2.2 · Application]

Illustrative figures.

A contractor recognises revenue over time on a construction contract using a cost-to-cost input method. The transaction price is USD 10,000,000 and the estimated total cost to complete is USD 8,000,000. Costs incurred to date are USD 3,000,000. What revenue should be recognised to date?

- **A.** USD 5,000,000
- **B.** USD 3,750,000
- **C.** USD 3,000,000
- **D.** USD 2,400,000

Key: B.

Progress is measured as costs incurred against estimated total costs, and that proportion is applied to the transaction price:

$$\text{Progress} = 3,000,000 \div 8,000,000 = 0.375 = 37.5 \% \text{ Revenue to date} = 0.375 \times 10,000,000 = 3,750,000$$

A assumes the contract is half complete because it feels half complete, ignoring the measured progress figure the stem supplies. The method is called an input method precisely because progress is measured, not estimated by impression.

C recognises revenue equal to cost incurred — a zero-margin position. That treatment exists in specific circumstances, but it is not the general cost-to-cost result, and applying it by default understates revenue on every profitable contract.

D measures progress against the wrong denominator: $3,000,000 \div 10,000,000 = 30 \%$, then applies that to the estimated total cost: $0.30 \times 8,000,000 = 2,400,000$. Two errors compounding — comparing cost to price, then applying a progress percentage to a cost base rather than to the price.

Item 3 [D3 · KA 3.2 · Analysis]

A sponsor asks for a cost estimate to ± 10 per cent accuracy for a scheme whose scope is defined by a concept layout and a site plan, and nothing else. The most defensible professional response is to:

- **A.** Apply a 10 per cent contingency to a bottom-up estimate to produce the range.
- **B.** Produce the ± 10 per cent figure and record the assumptions it rests on.
- **C.** Build a parametric model from the concept layout and present its output at ± 10 per cent.
- **D.** State that the scope definition available supports only an order-of-magnitude range.

Key: D.

Estimate accuracy is a property of the maturity of the scope definition and the estimating method used, not a service level that can be requested. The estimate classification systems in common use — those published by AACE International among them — tie accuracy ranges to the level of project

definition, and a concept layout sits at the least mature end of that scale. The professional answer names what the basis will support and sets out what scope definition an estimate of the requested accuracy would require, which turns a refusal into a plan.

A confuses contingency with accuracy. Contingency is a provision for identified risk within the estimate; accuracy is a statement about the estimate's own uncertainty. Adding a percentage to the number does not narrow the range around it, and presenting it as though it did is the mechanism by which under-defined estimates become approved budgets.

B is the option most people choose, and it fails for a specific reason: recording assumptions does not repair an accuracy claim the basis cannot support. The sponsor takes the ± 10 per cent into the funding paper; the assumptions stay in the appendix.

C substitutes a method for a basis. A parametric model built on concept-level inputs inherits the uncertainty of those inputs. The technique is sound and the accuracy claim is still unsupported.

Item 4 [D4 · KA 4.2 · Application]

Illustrative figures.

A work package budgeted 2,000 tonnes of reinforcement at USD 900 per tonne. Actual consumption was 2,100 tonnes at USD 880 per tonne. What are the price and quantity variances?

- **A.** Price USD 42,000 adverse; quantity USD 90,000 favourable.
- **B.** Price USD 90,000 adverse; quantity USD 42,000 favourable.
- **C.** Price USD 42,000 favourable; quantity USD 90,000 adverse.
- **D.** A single adverse variance of USD 48,000; the split cannot be derived from the data given.

Key: C.

Budget = $2,000 \times 900 = 1,800,000$ Actual = $2,100 \times 880 = 1,848,000$ Total variance = $1,848,000 - 1,800,000 = 48,000$ adverse

Price variance = (budget rate - actual rate) $\times$ actual quantity = $(900 - 880) \times 2,100 = 42,000$ favourable
Quantity variance = (budget quantity - actual quantity) $\times$ budget rate = $(2,000 - 2,100) \times 900 = 90,000$ adverse

Reconciliation: $90,000$ adverse - $42,000$ favourable = $48,000$ adverse

The commercial story is the point of the decomposition: buying was good and consumption was poor, and the headline 48,000 conceals both. Reporting only the net figure loses the two facts a manager can act on.

A and **B** invert the signs or swap the labels. Both are arithmetically derived from the same two magnitudes, which is deliberate — the item tests the decomposition, not the multiplication.

D claims the split is underivable. It is derivable: the stem supplies rate and quantity for both budget and actual, which is exactly the data the decomposition needs. Candidates who select D usually know the method and have not recognised that they already have everything required.

— 5. Group 2 — Project management principles (Domains 5-12)

Item 5 [D5 · KA 5.2 · Analysis]

Illustrative figures.

The month-end cost report shows actual cost of USD 8,600,000 against a plan of USD 9,000,000, and the project manager is reporting an under-run. The cost engineer establishes that USD 1,100,000 of goods were received during the month and have not been invoiced, and that a subcontract variation instructed during the month has not yet been valued, with a first estimate of USD 400,000. The most accurate statement of the position is:

- **A.** Cost incurred is USD 9,700,000 — USD 700,000 over plan — with a further USD 400,000 trended for the variation.
- **B.** Cost is USD 8,600,000 and the project is USD 400,000 under plan; uninvoiced receipts are not yet cost.
- **C.** Cost is USD 8,600,000 and will be restated when the invoices arrive, so no action is needed now.
- **D.** Cost is USD 10,100,000, because an instructed variation is incurred as soon as it is instructed.

Key: A.

$\text{Cost incurred} = 8,600,000 + 1,100,000 = 9,700,000$
 $\text{Variance to plan} = 9,700,000 - 9,000,000 = 700,000$
 $\text{over Forecast exposure including the trend} = 9,700,000 + 400,000 = 10,100,000$

The item turns on the commitment-accrual-actual cycle. Goods received are cost incurred; the invoice is a document, not an economic event. An instructed but unvalued variation is a **trend** — a known exposure carried in the forecast — and not yet a measured incurred cost.

B is the position the ledger reports if nobody looks, and it is wrong by USD 1,100,000. The report measures invoices and is being read as though it measured cost.

C is **B** with a delay attached, and it is worse, because it acknowledges the problem and defers it. The whole purpose of a cut-off is that a period's cost is stated in that period.

D over-corrects. Carrying an unvalued instruction as incurred cost overstates the ledger position and breaks the reconciliation to the accounts, which is the one thing a cost report must never do. A trend belongs in the forecast, visibly, not in the actuals.

Item 6 [D6 · KA 6.3 · Analysis]

Illustrative figures.

A project reports budget at completion (BAC) USD 12,000,000, earned value (EV) USD 4,500,000, actual cost (AC) USD 5,000,000 and planned value (PV) USD 5,400,000. Investigation establishes that the schedule slippage was caused by a permit delay that has now been resolved, while the cost inefficiency is a systemic labour productivity shortfall expected to continue. Which estimate at completion (EAC) is the most defensible forecast?

- **A.** USD 12,000,000
- **B.** USD 12,500,000
- **C.** USD 13,333,333

- **D.** USD 15,000,000

Key: C.

$$\text{CPI} = \text{EV} \div \text{AC} = 4,500,000 \div 5,000,000 = 0.90 \quad \text{SPI} = \text{EV} \div \text{PV} = 4,500,000 \div 5,400,000 = 0.8333$$

The cost inefficiency is systemic and is expected to continue, so past cost performance should be carried forward:

$$\text{EAC} = \text{BAC} \div \text{CPI} = 12,000,000 \div 0.90 = 13,333,333 \text{ (to the nearest dollar)}$$

A is the budget at completion. It is what was agreed, not what will happen. Reporting BAC as the forecast because the baseline has not been changed is one of the most common ways a project arrives at its overrun without warning anyone.

B is $\text{EAC} = \text{AC} + (\text{BAC} - \text{EV}) = 5,000,000 + 7,500,000 = 12,500,000$ — the atypical-variance form, which assumes remaining work will be performed at budgeted efficiency. It is the right method when the overrun has an identified, closed cause. Here the cause is systemic and continuing, so it understates.

D is $\text{EAC} = \text{AC} + (\text{BAC} - \text{EV}) \div (\text{CPI} \times \text{SPI}) = 5,000,000 + 7,500,000 \div 0.75 = 5,000,000 + 10,000,000 = 15,000,000$, carrying both cost and schedule inefficiency forward ($0.90 \times 0.8333 = 0.75$). That form is appropriate when schedule pressure is itself driving cost — overtime, acceleration, resequencing. Here the schedule driver has been removed, so applying it forecasts a penalty the project will not incur.

The assessed skill is not the arithmetic. All four figures are computable in under a minute. It is reading the situation to decide **which** inefficiency continues, and that is why the formula policy in EXB-01 – Examination blueprint – PCL-AI §7.2 provides no formula sheet: selecting the method is the assessment.

Item 7 [D7 · KA 7.4 · Application]

Illustrative figures.

A contract has a contract sum of USD 20,000,000 and provides for retention of 5 per cent, capped at 3 per cent of the contract sum. Work executed to date is valued at USD 3,400,000 gross. Previous certificates total USD 2,500,000 gross. What amount is due on this application, before tax?

- **A.** USD 730,000
- **B.** USD 855,000
- **C.** USD 900,000
- **D.** USD 3,230,000

Key: B.

First test the cap, because a candidate who does not is guessing that it does not bite:

$$\text{Retention cap} = 3 \% \times 20,000,000 = 600,000 \quad \text{Cumulative retention} = 5 \% \times 3,400,000 = 170,000 \text{ — below the cap, so 5 per cent applies in full.}$$

$$\begin{aligned} \text{Cumulative certified net} &= 3,400,000 - 170,000 = 3,230,000 \\ \text{Previously certified net} &= 2,500,000 \\ &- (5 \% \times 2,500,000) = 2,500,000 - 125,000 = 2,375,000 \\ \text{Due this application} &= 3,230,000 - 2,375,000 = 855,000 \end{aligned}$$

$$\begin{aligned} \text{Cross-check on the increment: gross increment} &= 3,400,000 - 2,500,000 = 900,000; \\ \text{retention on the increment} &= 5 \% \times 900,000 = 45,000; \\ &900,000 - 45,000 = 855,000 \end{aligned}$$

A deducts the cumulative retention of 170,000 from the incremental gross valuation of 900,000. Mixing a cumulative figure with an incremental one is the defining error of interim valuation, and it produces an under-certification that the contractor will find and dispute.

C ignores retention entirely and certifies the gross increment.

D is the cumulative net position — correct as a cumulative figure and wrong as an amount due, because it takes no account of what has already been certified and paid.

Item 8 [D8 · KA 8.4 · Recall]

The performance measurement baseline is revised:

- **A.** Whenever actual cost differs materially from planned cost.
- **B.** At each monthly reporting cycle, to reflect the latest forecast.
- **C.** When the forecast at completion exceeds the budget at completion.
- **D.** Only through approved change control, whatever the size of the variance.

Key: D.

A baseline is the agreed measurement reference. It changes only by an approved change, because variance is by definition the difference between performance and the baseline — if the baseline moves to meet performance, variance is structurally zero and the measurement has been destroyed.

A and **C** both re-baseline in response to a variance, which is precisely the circumstance in which the baseline must hold still. **B** describes rubber baselining, the practice of quietly re-setting the reference each month so that every report shows the project on plan. It is the most damaging thing that can be done to a controls function, and it is almost always done incrementally and with good intentions.

Approved change control revises a baseline for a scope, schedule or budget change that has been authorised. It does not revise a baseline because the project is behind.

Item 9 [D9 · KA 9.5 · Application]

Illustrative figures.

A product team of eight costs USD 120,000 per two-week sprint, fully loaded. The remaining backlog is estimated at 240 story points and is stable. The team's planned velocity is 30 points per sprint. Its observed velocities over the last five sprints were 34, 28, 31, 27 and 20 points; the most recent sprint fell across a public-holiday period. Sprints cannot be part-purchased. What is the most defensible forecast of remaining cost?

- **A.** USD 1,440,000
- **B.** USD 1,080,000
- **C.** USD 1,028,571
- **D.** USD 960,000

Key: B.

Average observed velocity = $(34 + 28 + 31 + 27 + 20) \div 5 = 140 \div 5 = 28$ points per sprint
 Sprints remaining = $240 \div 28 = 8.57$, rounded up to 9 whole sprints
 Remaining cost = $9 \times 120,000 = 1,080,000$

Two judgements sit inside that arithmetic, and both must be named alongside the answer. **Assumption one:** the observed five-sprint average is a better predictor than either the plan or the most recent sprint. **Assumption two:** the backlog is stable, as the stem states — if scope grows, the forecast moves, and in adaptive delivery scope growth is normal rather than exceptional.

A extrapolates from the most recent sprint alone: $240 \div 20 = 12$ sprints; $12 \times 120,000 = 1,440,000$. The stem tells you that sprint was depressed by a holiday period. Forecasting from a single known-atypical observation is the schedule-forecasting equivalent of re-baselining to the worst month.

C treats sprint capacity as divisible: $8.57 \times 120,000 = 1,028,571$. You cannot buy 0.57 of a sprint. The team is paid for whole sprints, and rounding down hides a period of cost.

D forecasts from the plan rather than from performance: $240 \div 30 = 8$ sprints; $8 \times 120,000 = 960,000$. Substituting the plan for observed performance is the same error as reporting BAC as the forecast in Item 6, in a different vocabulary — and it is more tempting here, because the planned velocity is a published team commitment.

Item 10 [D10 · KA 10.2 · Analysis]

Illustrative figures.

A network has two paths from a common start to a common finish. Activity A (5 days) precedes both B (7 days) and C (3 days). C precedes D (6 days). B and D both precede E (4 days), which finishes the project. Activity B is delayed by 3 days. What is the effect?

- **A.** The project finishes one day late, and the critical path becomes A-B-E.
- **B.** No effect: activity B has two days of total float.
- **C.** The project finishes two days late, and the critical path is unchanged.
- **D.** The project finishes three days late, since the whole delay flows to the finish.

Key: A.

Path A-B-E = $5 + 7 + 4 = 16$ days Path A-C-D-E = $5 + 3 + 6 + 4 = 18$ days

The critical path is A-C-D-E at 18 days, and B carries total float of $18 - 16 = 2$ days.

After the 3-day delay:

Path A-B-E = $5 + 10 + 4 = 19$ days Path A-C-D-E = 18 days (unchanged) Project duration = 19 days — one day later than the 18-day baseline

Float absorbs the first two days. The third day pushes the path past 18, the project finishes one day late, and **the critical path moves to A-B-E.**

B is right up to a point and stops too early — it applies the float and does not check whether the delay exceeds it. This is the most instructive wrong answer in the set: float is a quantity that gets consumed, not a permission.

C subtracts wrongly and misses the path switch. **D** ignores float altogether and passes the whole delay to the finish date, which is what a spreadsheet does when nobody has modelled the network.

The arithmetic here is trivial by design. The item is about whether you reason over a network, and about remembering that a critical path is a property of the current schedule, not a label attached to activities at the start of the job.

Item 11 [D11 · KA 11.3 · Analysis]

In a four-person project office, the same cost engineer raises purchase requisitions, records goods receipts in the enterprise resource planning system, and approves supplier invoices for payment up to a threshold. The control weakness this creates is best described as:

- **A.** A data-quality risk, since one person's coding conventions are never independently checked.
- **B.** A capacity risk, since the cost engineer becomes a single point of failure during absence.
- **C.** One person controls initiation, receipt and approval, so a fictitious purchase could be paid unseen.
- **D.** An acceptable efficiency in a small team, provided the approval threshold is set low.

Key: C.

Segregation of duties exists so that no single person can both initiate and complete a transaction that moves money. Here the same individual controls all three stages of the procure-to-pay cycle, so a fictitious or inflated purchase could be requisitioned, receipted and approved without any second party seeing it. That is the control weakness; everything else is a consequence or an unrelated risk.

A and **B** describe genuine risks that this arrangement also creates. Neither is the control weakness, and a candidate who selects either has identified a real problem and answered a different question.

D is the option that most deserves examining, because it is half right in a way that matters in practice. Perfect segregation is genuinely impossible in a four-person office, and pretending otherwise produces a control framework nobody follows. But a low threshold is a **compensating control**, not the absence of a weakness — and a compensating control only counts when it is documented, operated and monitored. The professional response to a small team is to name the weakness, then put in place compensating controls that can be evidenced: independent review of a sample of transactions, and exception reporting that lists every case where requisition, receipt and approval share a user identity. Accepting the arrangement without either is the answer D actually describes.

Item 12 [D12 · KA 12.2 · Analysis]

Illustrative figures.

A single risk, if it occurs, would delay handover by four months and trigger liquidated damages of USD 4,000,000. The probability is assessed at 15 per cent. The finance director proposes carrying USD 600,000 of contingency for it, being $0.15 \times 4,000,000$. The most defensible controls response is to:

- **A.** Carry USD 600,000 as contingency, since expected monetary value is the standard basis for quantifying risk.
- **B.** Carry USD 4,000,000 as contingency, since the outcome is binary and the project must absorb it.
- **C.** Carry nothing, since liquidated damages are a contractual remedy and not a project cost risk.
- **D.** Report the exposure as 0 or USD 4,000,000 at 15 per cent likelihood, and set the reserve by recorded management decision.

Key: D.

Expected monetary value = $0.15 \times 4,000,000 = 600,000$ — and the project will never experience USD 600,000 of this risk. It will experience nothing or USD 4,000,000.

Expected monetary value is a long-run average. It is a sound basis for aggregating a portfolio of many independent risks, where the averaging has something to average over. Applied to a single dominant binary risk it produces a number that is arithmetically correct and describes no possible outcome, and it conceals the decision that actually has to be taken: how much of a 15 per cent chance of a USD 4,000,000 hit the organisation is willing to carry unfunded. That is a risk-appetite decision for management, and the controls professional's job is to present the exposure so that it can be taken knowingly and recorded.

A is not an arithmetic error; it is a modelling error, and it is extremely common because the calculation is easy and the number looks responsible in a report.

B is not wrong in principle — fully funding a severe binary exposure is a legitimate choice. It is not the most defensible response, because it is a decision the controls professional does not have the standing to take unilaterally, and because reserving the full amount by default freezes capital that the organisation may rationally prefer to put at risk. This is a single-best-answer item in the strict sense: **B** is permissible, **D** is what a competent professional does first.

C excludes a quantifiable cost consequence on the grounds that its origin is contractual. Liquidated damages hit the project's cost outturn like any other cost.

— 6. Group 3 — AI knowledge & practical approach (Domain 13)

Item 13 [D13 · KA 13.1 · Recall]

Which statement about the output of a large language model (LLM) is correct?

- **A.** It is retrieved from a verified source, so it can be cited directly.
- **B.** It is generated as a likely continuation, so it can be fluent and wrong.
- **C.** It is deterministic: an identical prompt always returns identical text.
- **D.** It carries a reliability score indicating whether the content is correct.

Key: B.

A large language model produces text by generating statistically likely continuations, not by retrieving checked statements from a store. Fluency and correctness are therefore **independent properties of the output** — which is the single most consequential fact for any professional who relies on one, because human readers use fluency as a proxy for competence and the proxy does not hold here.

A describes retrieval. A generative model can produce a citation-shaped string that refers to nothing at all, and it will produce it with exactly the same confidence as a real one.

C is false in general. Most deployments sample from the distribution, so repeated identical prompts can return different text — a genuine problem for auditability, and one reason a deterministic rule is preferred wherever the logic is known (see Item 15). Even where sampling is disabled, reproducibility is not correctness.

D conflates two different quantities. Some interfaces expose token-level probabilities, but the probability that a token is a likely continuation is not the probability that a statement is true. A confident model and a correct model are not the same thing.

Item 14 [D13 · KA 13.2 · Analysis]

A controls team wants to train a model to predict cost overrun at control-account level using five years of historical project data. The cost ledger has been recoded twice in that period, and neither earlier coding structure has been mapped to the current one. The most important consequence is:

- **A.** Training will take longer, because the model must reconcile three coding structures.
- **B.** The model will favour recent projects, which is corrected by adding more historical data.
- **C.** Features and labels mean different things across the period, so the model learns from incompatible definitions.
- **D.** The model cannot be built until the whole ledger history has been recoded to the current structure.

Key: C.

The same field name refers to different things in different years. A model trained across that history learns from a mixture of incompatible definitions, and — this is the part that makes it dangerous rather than merely disappointing — **a held-out test set drawn from the same inconsistent history will report respectable accuracy**. The failure is silent. It surfaces only when the model is used on live data and produces confident predictions that nobody can explain.

A treats a validity problem as a performance problem. Training time is irrelevant.

B misdiagnoses the mechanism and then prescribes more of the poison: adding further data from the same inconsistently coded history does not repair a definition problem, it enlarges it.

D overstates. Recoding the entire history is one remedy and rarely the practical one. Restricting training to the consistently coded period, or building an explicit mapping between structures and validating it on a sample, are both legitimate. The professional response is to establish what the data means before modelling it, and to document that establishment — which is the whole of knowledge area 13.2 in one sentence.

Item 15 [D13 · KA 13.4 · Analysis]

Each week, a controls team wants to flag every control account whose cost performance index has fallen by more than a published threshold since the previous period, and to issue a short written explanation of each flag. The best-governed division of labour is:

- **A.** A deterministic rule detects the deterioration; a generative model drafts the explanation; the engineer verifies both.
- **B.** A generative model both detects the deterioration against the threshold and drafts the explanation.
- **C.** A generative model detects the deterioration and the cost engineer writes each explanation.
- **D.** A model is trained on which accounts cost engineers have flagged before, and flags accordingly.

Key: A.

Match the technique to the task. Detection here is a **defined, auditable rule**: compute an index, compare the change against a published threshold, flag. That is deterministic logic, it is reproducible, and it can be reconciled — run it twice and it gives the same answer, and anyone can check why a particular account was flagged. Explanation is a **language task**, where a generative model genuinely helps, provided a competent human owns the output before it is issued. The verification step is not a formality; it is the step that makes the whole arrangement defensible.

B hands a deterministic test to a probabilistic system. It buys opacity and non-reproducibility for nothing: two runs may disagree, and the flag list cannot be reconciled to the underlying figures. Whenever a rule will do, a rule is better — cheaper, faster, auditable and explainable.

C allocates both tasks to the participant worst suited to each: the model to the reproducible test, the human to the repetitive drafting. It is the mirror image of the right answer.

D solves a problem nobody has. Training a model to reproduce what cost engineers have flagged before encodes past human behaviour, including its inconsistencies and its blind spots, in place of a published threshold that already exists and that anyone can read. It also makes the flag list impossible to explain to a sponsor, since the answer to "why was this account flagged?" becomes "because the model learned that we usually flag accounts like this".

Item 16 [D13 · KA 13.6 · Analysis]

A generative-AI assistant drafts the cost narrative for the monthly report. The cost manager reviews it, corrects two figures, and issues the report. For the audit trail, the minimum that must be recorded is:

- **A.** A statement in the report that AI tools were used in its preparation.
- **B.** Nothing: the cost manager issued the report and is accountable for it.
- **C.** A per-paragraph confidence score reported by the model alongside the draft.
- **D.** The prompt, the model, the output as generated, the corrections and the approver.

Key: D.

Auditability means being able to reconstruct, afterwards, what the tool produced, what a human changed, and who owned the result. All four elements are needed: without the output as generated you cannot see what was corrected, and without the approver you cannot see who took responsibility.

A is a disclaimer, not a trail. It tells a reader that a tool was involved and nothing about what was checked, which is precisely the information a reviewer needs.

B is the option that deserves the most attention, because its premise is correct and its conclusion is wrong. Accountability does sit with the cost manager — "AI proposes; the professional disposes" is not negotiable. But accountability that cannot be evidenced cannot be demonstrated when it is questioned six months later, and demonstrating it is the entire purpose of an audit trail. A professional who is accountable and has no record is exposed, not protected.

C substitutes a machine's self-report for human review. A high confidence score attached to a fluent paragraph is still an unverified paragraph, and confidence is not correctness (see Item 13).

— 7. Answer key

Item	Key	Item	Key	Item	Key	Item	Key
1	A	5	A	9	B	13	B
2	B	6	C	10	A	14	C
3	D	7	B	11	C	15	A
4	C	8	D	12	D	16	D

— 8. Reading the items as an item writer

Publishing the rationale is useful. Publishing what the item is doing is more useful, and almost nobody does it. Four observations, keyed to the items above.

Every distractor is an arrival point, not a decoy. In Item 9, each of the three wrong answers is the figure you actually get by making one specific decision: extrapolating from a single atypical sprint, treating sprint capacity as divisible, or forecasting from the plan instead of from performance. A candidate who lands on one of them has not been tricked — they have completed a calculation, and the option they selected identifies exactly which judgement they made. That is what makes the pattern of a candidate's wrong answers diagnostic, and it is the standard [EXB-04 – How items are written, reviewed and retired](#) §4 holds authors to.

The arithmetic is deliberately easy where the judgement is hard. In Item 6, all four estimates at completion are computable in a minute; in Item 10, the path sums are trivial. The difficulty is entirely in deciding which inefficiency continues, and in remembering that float is consumed. An item that hides its assessment behind long multiplication is measuring stamina.

The key must be the best answer, not the only permissible one. Item 12 is the clearest case: fully reserving a severe binary exposure is a legitimate choice, and the item says so in the rationale rather than pretending option B is foolish. Candidates who argue with such an item instead of answering it lose the mark; candidates who ask "which of these would I do first" find it.

Option length is a cue, and judgement items fight it. The defensible answer in an Analysis item usually needs more words than a wrong one, because it carries a qualification the wrong answers do not. Item 12's key is the longest option in that set, which is a weakness the reviewers accepted rather than concealed — the alternative was to pad the distractors, which would have made three options less plausible to buy symmetry. Where you can see the tension in these items, you are seeing an honest trade-off rather than an oversight, and a candidate who has learned to notice option length has learned something that will help them on a badly written examination somewhere else.

— 9. How candidates get these wrong

Reading the options before establishing the position. The single most expensive habit. Items 5, 6 and 10 all become straightforward once you have written down what the data actually says, and all three become a guess between plausible alternatives if you have not.

Doing the arithmetic and stopping. Item 6 punishes this precisely: computing all four estimates at completion correctly and then choosing without deciding which inefficiency continues.

Answering the question you know instead of the one asked. Item 11 offers two real risks that are not the control weakness. Item 7 offers a figure that is correct as a cumulative position and wrong as an amount due.

Applying a method outside its assumptions. Item 12 is expected monetary value used where averaging has nothing to average over. Item 3 is an accuracy claim the basis cannot support.

Treating "the professional answer" as the cautious answer. It is not always. In Item 12 the cautious option — reserve the full amount — is not the key, because the decision is not the controls professional's to take alone.

Accepting a machine output because it reads well. Items 13, 14 and 16 all turn on the same point in different vocabulary. Fluency, confidence and accuracy are three different things, and only the third matters.

— 10. Checklist for working practice items

- [] I answered before reading the rationale, and wrote down my intermediate figure or position.
 - [] For every item I got wrong, I identified which distractor I chose and which named error it represents.
 - [] For every item I got right, I checked my reasoning against the rationale rather than my letter against the key.
 - [] I worked at least one set under time pressure rather than at leisure.
 - [] I noted which domains my errors cluster in, and adjusted study time using the weighting arithmetic in [EXB-02 §3](#).
 - [] I have not tried to memorise these items, which are study material and will not appear on any form.
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Publishing sixteen items with every distractor explained gives away nothing an examination needs to keep, and it settles a question a candidate would otherwise have to take on trust: whether "scenario-based" and "assesses judgement" describe the items or the marketing. Work them, and the claim in [EXB-01 – Examination blueprint – PCL-AI](#) becomes something you can check rather than something you are asked to believe.

— Related

- [EXB-01 – Examination blueprint – PCL-AI](#) — the sampling frame these items are written against, and the formula policy Item 6 depends on
 - [EXB-03 – Cognitive levels and item types](#) — what Recall, Application and Analysis mean, and the limits of the format
 - [EXB-07 – Examination security and integrity](#) — why published items and live items are permanently separate
 - [CER-01 – Certification handbook – master](#) — eligibility, booking, conduct and results
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— Sources and standards

All sixteen items are original, written for this document by PCI Editorial and reviewed for technical correctness. None is drawn from, adapted from, or otherwise related to any live examination item, from PCI or any other body. All figures are illustrative and were independently recomputed before submission; no item describes a real project, organisation or person.

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